Fluidum: the derivations of the generated matrices

Each section is written by the derivation script it describes, from the matrices, sources and check results that script computed:

- `deriv/derive_2p1d_viscous.wls`
- `deriv/derive_hq_2p1d.wls`
- `deriv/derive_hq_consistent_2p1d.wls`
- `deriv/derive_two_way.wls`

Regenerate with `deriv/render_all.sh`.

Conventions: boost-invariant Milne coordinates; every system is written source on the left, $A_t\,\partial_\tau\phi + A_x\,\partial_x\phi + A_y\,\partial_y\phi + S = 0$; only nonzero entries are listed; $u^\tau = \sqrt{1+u_\perp^2}$ is written for the root wherever it occurs.

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1 Viscous hydrodynamics: Israel–Stewart with $\delta_{\pi\pi}$

Script: `deriv/derive_2p1d_viscous.wls`

Writes: `src/Matrix/viscous_generated.jl`

1.1 Setup

Milne coordinates, boost invariant. Two charts:

$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2)$$
$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2)$$

$$T^{\mu\nu} = (e + p + \Pi)\, u^\mu u^\nu + (p + \Pi)\, g^{\mu\nu} + \pi^{\mu\nu}, \qquad e = T\, p'(T) - p(T)$$
$$\nabla_\mu T^{\mu\nu} = 0$$
$$\tau_\pi\, \Delta^{\mu\nu}_{\alpha\beta}\, D\pi^{\alpha\beta} + \pi^{\mu\nu} + 2\eta\, \sigma^{\mu\nu} + \delta_{\pi\pi}\, \pi^{\mu\nu}\theta = 0$$
$$\tau_\Pi\, D\Pi + \Pi + \zeta\, \theta = 0$$

with

$$D = u^\mu \nabla_\mu, \qquad \theta = \nabla_\mu u^\mu, \qquad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$
$$\sigma^{\mu\nu} = \Delta^{\mu\nu}_{\alpha\beta} \nabla^\alpha u^\beta, \qquad \Delta^{\mu\nu}_{\alpha\beta} = \tfrac{1}{2}\left(\Delta^\mu_\alpha \Delta^\nu_\beta + \Delta^\mu_\beta \Delta^\nu_\alpha\right) - \tfrac{1}{3}\Delta^{\mu\nu}\Delta_{\alpha\beta}$$

η, τ_π, ζ, τ_Π are not differentiated. Quasi-linear form, source on the left:

$$A_t\, \partial_\tau \phi + A_r\, \partial_r \phi + S = 0 \quad \text{(1+1D)}$$
$$A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0 \quad \text{(2+1D)}$$

The $\delta_{\pi\pi}$ term is linear, so each matrix and the source split as

$$A = A^{\mathrm{IS}} + \delta_{\pi\pi}\, A^{(\delta)}, \qquad S = S^{\mathrm{IS}} + \delta_{\pi\pi}\, S^{(\delta)}$$

1.2 1+1D cylindrical

Fields, with π^{rr} and the τ row of $\pi^{\mu\nu}$ fixed by $u_\mu \pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$:

$$\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi), \qquad \pi^\phi{}_\phi = r^2 \pi^{\phi\phi}, \quad \pi^\eta{}_\eta = \tau^2 \pi^{\eta\eta}$$
$$\pi^{rr} = -\left((u^\tau)^2\, (\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right), \qquad u^\tau = \sqrt{(u^r)^2 + 1}$$

Rows: $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu r}$, $E^{\phi\phi}$, $E^{\eta\eta}$, E_Π, where $E^{\mu\nu}$ and E_Π are the left-hand sides of the shear and bulk equations.

1.2.1 IS part

$$(A_t^{\mathrm{IS}})_{11} = (u^r)^2\, (p''(T)T + p'(T)) + p''(T)T$$
$$(A_t^{\mathrm{IS}})_{12} = 2u^r (p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi)$$
$$(A_t^{\mathrm{IS}})_{13} = -(u^r)^2$$
$$(A_t^{\mathrm{IS}})_{14} = -(u^r)^2$$
$$(A_t^{\mathrm{IS}})_{15} = (u^r)^2$$
$$(A_t^{\mathrm{IS}})_{21} = u^r u^\tau (p''(T)T + p'(T))$$

$$
\begin{aligned}
(A_t^{\mathrm{IS}})_{22} &= \frac{\left(2\left(u^r\right)^2+1\right)\left(p'(T)T+\Pi-{\pi^\eta}_\eta-{\pi^\phi}_\phi\right)}{u^\tau}\\
(A_t^{\mathrm{IS}})_{23} &= -u^r u^\tau\\
(A_t^{\mathrm{IS}})_{24} &= -u^r u^\tau\\
(A_t^{\mathrm{IS}})_{25} &= u^r u^\tau\\
(A_t^{\mathrm{IS}})_{32} &= -\frac{2u^r\eta}{3r^2u^\tau}\\
(A_t^{\mathrm{IS}})_{33} &= \frac{\tau_\pi u^\tau}{r^2}\\
(A_t^{\mathrm{IS}})_{42} &= -\frac{2u^r\eta}{3\tau^2u^\tau}\\
(A_t^{\mathrm{IS}})_{44} &= \frac{\tau_\pi u^\tau}{\tau^2}\\
(A_t^{\mathrm{IS}})_{52} &= \frac{u^r\zeta}{u^\tau}\\
(A_t^{\mathrm{IS}})_{55} &= \tau_\Pi u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_r^{\mathrm{IS}})_{11} &= u^r u^\tau(p''(T)T+p'(T))\\
(A_r^{\mathrm{IS}})_{12} &= \frac{\left(2\left(u^r\right)^2+1\right)\left(p'(T)T+\Pi-{\pi^\eta}_\eta-{\pi^\phi}_\phi\right)}{u^\tau}\\
(A_r^{\mathrm{IS}})_{13} &= -u^r u^\tau\\
(A_r^{\mathrm{IS}})_{14} &= -u^r u^\tau\\
(A_r^{\mathrm{IS}})_{15} &= u^r u^\tau\\
(A_r^{\mathrm{IS}})_{21} &= p''(T)\left(u^r\right)^2T+p'(T)\left(u^r\right)^2+p'(T)\\
(A_r^{\mathrm{IS}})_{22} &= 2u^r(p'(T)T+\Pi-{\pi^\eta}_\eta-{\pi^\phi}_\phi)\\
(A_r^{\mathrm{IS}})_{23} &= -\left(u^r\right)^2-1\\
(A_r^{\mathrm{IS}})_{24} &= -\left(u^r\right)^2-1\\
(A_r^{\mathrm{IS}})_{25} &= \left(u^\tau\right)^2\\
(A_r^{\mathrm{IS}})_{32} &= -\frac{2\eta}{3r^2}\\
(A_r^{\mathrm{IS}})_{33} &= \frac{\tau_\pi u^r}{r^2}\\
(A_r^{\mathrm{IS}})_{42} &= -\frac{2\eta}{3\tau^2}\\
(A_r^{\mathrm{IS}})_{44} &= \frac{\tau_\pi u^r}{\tau^2}\\
(A_r^{\mathrm{IS}})_{52} &= \zeta\\
(A_r^{\mathrm{IS}})_{55} &= \tau_\Pi u^r
\end{aligned}
$$

$$
\begin{aligned}
S_1^{\mathrm{IS}} &= \frac{1}{r\tau u^\tau}\Big(\left(u^\tau\right)^2\left(p'(T)T+\Pi\right)(ru^\tau+\tau u^r)-{\pi^\phi}_\phi u^r\left(ru^ru^\tau+\tau\left(u^r\right)^2+\tau\right)\\
&\quad-{\pi^\eta}_\eta\left(r\left(\left(u^r\right)^2-1\right)u^\tau+\tau\left(\left(u^r\right)^3+u^r\right)\right)\Big)\\
S_2^{\mathrm{IS}} &= \frac{1}{r\tau u^\tau}\Big(-{\pi^\eta}_\eta\left(u^\tau\right)^2(ru^r+\tau u^\tau)+u^r\left(p'(T)T+\Pi\right)\left(r\left(u^r\right)^2+r+\tau u^ru^\tau\right)\\
&\quad-{\pi^\phi}_\phi\left(r\left(\left(u^r\right)^3+u^r\right)+\tau\left(\left(u^r\right)^2+2\right)u^\tau\right)\Big)\\
S_3^{\mathrm{IS}} &= \frac{{\pi^\phi}_\phi}{r^2}-\frac{2\eta\left(r\left(u^r\right)^2+r-2\tau u^ru^\tau\right)}{3r^3\tau u^\tau}\\
S_4^{\mathrm{IS}} &= \frac{3{\pi^\eta}_\eta\tau-\frac{2\tau u^r\eta}{r}+4u^\tau\eta}{3\tau^3}\\
S_5^{\mathrm{IS}} &= \Pi+\frac{u^r\zeta}{r}+\frac{u^\tau\zeta}{\tau}
\end{aligned}
$$

1.2.2 $\delta_{\pi\pi}$ part

$$(A_t^{(\delta)})_{32} = \frac{\pi^{\phi}{}_{\phi} u^r}{r^2 u^\tau}$$
$$(A_t^{(\delta)})_{42} = \frac{\pi^{\eta}{}_{\eta} u^r}{\tau^2 u^\tau}$$

$$(A_r^{(\delta)})_{32} = \frac{\pi^{\phi}{}_{\phi}}{r^2}$$
$$(A_r^{(\delta)})_{42} = \frac{\pi^{\eta}{}_{\eta}}{\tau^2}$$

$$S_3^{(\delta)} = \frac{\pi^{\phi}{}_{\phi}(r u^\tau + \tau u^r)}{r^3 \tau}$$
$$S_4^{(\delta)} = \frac{\pi^{\eta}{}_{\eta}(r u^\tau + \tau u^r)}{r \tau^3}$$

1.3 2+1D Cartesian

Fields, with π^{xx} and the τ row fixed by the same constraints:

$$\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \Pi)$$
$$\pi^{xx} = -\frac{\pi^{\eta}{}_{\eta} \left(u^\tau\right)^2 - 2\pi^{xy} u^x u^y + \pi^{yy} \left(u^x\right)^2 + \pi^{yy}}{\left(u^y\right)^2 + 1}, \qquad u^\tau = \sqrt{\left(u^x\right)^2 + \left(u^y\right)^2 + 1}$$

Rows: $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu x}$, $\nabla_\mu T^{\mu y}$, E^{yy}, $E^{\eta\eta}$, E^{xy}, E_Π.

1.3.1 IS part

$$(A_t^{\mathrm{IS}})_{11} = \left(u^\tau\right)^2 \left(p''(T)T + p'(T)\right) - p'(T)$$
$$(A_t^{\mathrm{IS}})_{12} = \frac{2\left(p'(T)T u^x \left(u^y\right)^2 + p'(T)T u^x + \Pi u^x \left(\left(u^y\right)^2 + 1\right) - \pi^{\eta}{}_{\eta} u^x + \pi^{xy} u^y - \pi^{yy} u^x\right)}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{13} = \frac{2}{\left(\left(u^y\right)^2 + 1\right)^2}\Big(\pi^{xy}\left(u^x - u^x \left(u^y\right)^2\right)$$
$$+ u^y\Big(\pi^{yy} + p'(T)T + \pi^{\eta}{}_{\eta}\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + 2p'(T)T\left(u^y\right)^2 + p'(T)T\left(u^y\right)^4$$
$$+ \Pi\left(\left(u^y\right)^2 + 1\right)^2\Big)\Big)$$
$$(A_t^{\mathrm{IS}})_{14} = \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{15} = -\frac{\left(u^x\right)^2}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{16} = \frac{2u^x u^y}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{17} = \left(u^x\right)^2 + \left(u^y\right)^2$$
$$(A_t^{\mathrm{IS}})_{21} = u^\tau u^x (p''(T)T + p'(T))$$
$$(A_t^{\mathrm{IS}})_{22} = \frac{1}{\left(u^\tau\right)^3}\Big(\left(u^\tau\right)^4\left(p'(T)T + \Pi\right) + \left(u^\tau\right)^2\left(u^x\right)^2\left(p'(T)T + \Pi\right)$$
$$+ \frac{u^x\left(\pi^{\eta}{}_{\eta}\left(u^\tau\right)^2 u^x - \pi^{xy}u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \pi^{yy}\left(\left(u^x\right)^3 + u^x\right)\right)}{\left(u^y\right)^2 + 1}$$

$$
\begin{aligned}
&- \frac{(u^\tau)^2 \left(\pi^\eta{}_\eta \left(3\,(u^x)^2 + (u^y)^2 + 1\right) - 4\pi^{xy} u^x u^y + 3\pi^{yy}\,(u^x)^2 + \pi^{yy}\right)}{(u^y)^2 + 1}\Big) \\
(A_t^{\mathrm{IS}})_{23} &= \frac{1}{(u^\tau)^3}\Big((u^\tau)^2\, u^x u^y (p'(T)T + \Pi) + u^y \left(\frac{u^x \left(\pi^\eta{}_\eta\,(u^\tau)^2 - 2\pi^{xy} u^x u^y + \pi^{yy}\,(u^x)^2 + \pi^{yy}\right)}{(u^y)^2 + 1} - \pi^{xy} u^y\right) \\
&+ \frac{(u^\tau)^2 \left(2u^x u^y \left(\pi^\eta{}_\eta\,(u^x)^2 + \pi^{yy}\,(u^x)^2 + \pi^{yy}\right) + \pi^{xy}\left(\left((u^y)^2 + 1\right)^2 - 2\,(u^x)^2\left((u^y)^2 - 1\right)\right)\right)}{\left((u^y)^2 + 1\right)^2}\Big) \\
(A_t^{\mathrm{IS}})_{24} &= -\frac{u^x \left((u^x)^2 + 1\right)}{u^\tau \left((u^y)^2 + 1\right)} \\
(A_t^{\mathrm{IS}})_{25} &= -\frac{u^\tau u^x}{(u^y)^2 + 1} \\
(A_t^{\mathrm{IS}})_{26} &= \frac{u^y \left(\frac{2(u^x)^2}{(u^y)^2+1} + 1\right)}{u^\tau} \\
(A_t^{\mathrm{IS}})_{27} &= u^\tau u^x \\
(A_t^{\mathrm{IS}})_{31} &= u^\tau u^y (p''(T)T + p'(T)) \\
(A_t^{\mathrm{IS}})_{32} &= \frac{u^x u^y \left(p'(T)T\,(u^\tau)^2 + \Pi\,(u^\tau)^2 - \pi^{yy}\right) + \pi^{xy}\left((u^y)^2 + 1\right)}{(u^\tau)^3} \\
(A_t^{\mathrm{IS}})_{33} &= \frac{(u^\tau)^4 (p'(T)T + \Pi) + (u^\tau)^2 (u^y)^2 (p'(T)T + \Pi) - u^y(\pi^{xy} u^x + \pi^{yy} u^y) + \pi^{yy}\,(u^\tau)^2}{(u^\tau)^3} \\
(A_t^{\mathrm{IS}})_{34} &= \frac{u^y}{u^\tau} \\
(A_t^{\mathrm{IS}})_{36} &= \frac{u^x}{u^\tau} \\
(A_t^{\mathrm{IS}})_{37} &= u^\tau u^y \\
(A_t^{\mathrm{IS}})_{42} &= -\frac{2\left(3\pi^{xy}\tau_\pi u^y \left((u^y)^2 + 1\right) + u^x \left((u^y)^2\,(\eta - 3\pi^{yy}\tau_\pi) + \eta\right)\right)}{3u^\tau} \\
(A_t^{\mathrm{IS}})_{43} &= \frac{2u^y \left(3\pi^{xy}\tau_\pi u^x u^y - 3\pi^{yy}\tau_\pi \left((u^x)^2 + 1\right) + \eta\left(3\,(u^x)^2 + 2\,(u^y)^2 + 2\right)\right)}{3u^\tau} \\
(A_t^{\mathrm{IS}})_{44} &= \tau_\pi u^\tau \\
(A_t^{\mathrm{IS}})_{52} &= -\frac{2u^x \eta}{3\tau^2 u^\tau} \\
(A_t^{\mathrm{IS}})_{53} &= -\frac{2u^y \eta}{3\tau^2 u^\tau} \\
(A_t^{\mathrm{IS}})_{55} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t^{\mathrm{IS}})_{62} &= \frac{1}{3u^\tau}\Big(-3\pi^{xy}\tau_\pi u^x \left(2\,(u^y)^2 + 1\right) \\
&+ u^y \left(3\pi^\eta{}_\eta \tau_\pi\,(u^\tau)^2 + 3\pi^{yy}\left(2\tau_\pi\,(u^x)^2 + \tau_\pi\right) + \eta\left((u^x)^2 + 3\,(u^y)^2 + 3\right)\right)\Big) \\
(A_t^{\mathrm{IS}})_{63} &= \frac{1}{3u^\tau\left((u^y)^2 + 1\right)}\Big(-3\pi^{yy}\tau_\pi u^x \left((u^x)^2 + 1\right)\left(2\,(u^y)^2 + 1\right) + 3\pi^{xy}\tau_\pi u^y \left(\left(2\,(u^x)^2 - 1\right)(u^y)^2 - 1\right) \\
&+ u^x \left(\left((u^y)^2 + 1\right)\eta\left(3\,(u^x)^2 + (u^y)^2 + 3\right) - 3\pi^\eta{}_\eta \tau_\pi\,(u^\tau)^2\,(u^y)^2\right)\Big) \\
(A_t^{\mathrm{IS}})_{66} &= \tau_\pi u^\tau \\
(A_t^{\mathrm{IS}})_{72} &= \frac{u^x \zeta}{u^\tau} \\
(A_t^{\mathrm{IS}})_{73} &= \frac{u^y \zeta}{u^\tau} \\
(A_t^{\mathrm{IS}})_{77} &= \tau_\Pi u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_x^{\mathrm{IS}})_{11} &= u^\tau u^x (p''(T)T + p'(T)) \\
(A_x^{\mathrm{IS}})_{12} &= \frac{1}{(u^\tau)^3}\Big((u^\tau)^4 (p'(T)T + \Pi) + (u^\tau)^2 (u^x)^2 (p'(T)T + \Pi) \\
&\quad + \frac{u^x \left(\pi^\eta{}_\eta (u^\tau)^2 u^x - \pi^{xy} u^y \left(2(u^x)^2 + (u^y)^2 + 1 \right) + \pi^{yy} \left((u^x)^3 + u^x \right) \right)}{(u^y)^2 + 1} \\
&\quad - \frac{(u^\tau)^2 \left(\pi^\eta{}_\eta \left(3(u^x)^2 + (u^y)^2 + 1 \right) - 4\pi^{xy} u^x u^y + 3\pi^{yy} (u^x)^2 + \pi^{yy} \right)}{(u^y)^2 + 1} \Big) \\
(A_x^{\mathrm{IS}})_{13} &= \frac{1}{(u^\tau)^3}\Big((u^\tau)^2 u^x u^y (p'(T)T + \Pi) + u^y \left(\frac{u^x \left(\pi^\eta{}_\eta (u^\tau)^2 - 2\pi^{xy} u^x u^y + \pi^{yy} (u^x)^2 + \pi^{yy} \right)}{(u^y)^2 + 1} - \pi^{xy} u^y \right) \\
&\quad + \frac{(u^\tau)^2 \left(2u^x u^y \left(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy} (u^x)^2 + \pi^{yy} \right) + \pi^{xy} \left(\left((u^y)^2 + 1 \right)^2 - 2(u^x)^2 \left((u^y)^2 - 1 \right) \right) \right)}{\left((u^y)^2 + 1 \right)^2} \Big) \\
(A_x^{\mathrm{IS}})_{14} &= -\frac{u^x \left((u^x)^2 + 1 \right)}{u^\tau \left((u^y)^2 + 1 \right)} \\
(A_x^{\mathrm{IS}})_{15} &= -\frac{u^\tau u^x}{(u^y)^2 + 1} \\
(A_x^{\mathrm{IS}})_{16} &= \frac{u^y \left(\frac{2(u^x)^2}{(u^y)^2 + 1} + 1 \right)}{u^\tau} \\
(A_x^{\mathrm{IS}})_{17} &= u^\tau u^x \\
(A_x^{\mathrm{IS}})_{21} &= p''(T) T (u^x)^2 + p'(T) (u^x)^2 + p'(T) \\
(A_x^{\mathrm{IS}})_{22} &= 2u^x (p'(T)T + \Pi) - \frac{2(\pi^\eta{}_\eta u^x - \pi^{xy} u^y + \pi^{yy} u^x)}{(u^y)^2 + 1} \\
(A_x^{\mathrm{IS}})_{23} &= \frac{2 \left(u^y \left(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy} (u^x)^2 + \pi^{yy} \right) + \pi^{xy} \left(u^x - u^x (u^y)^2 \right) \right)}{\left((u^y)^2 + 1 \right)^2} \\
(A_x^{\mathrm{IS}})_{24} &= -\frac{(u^x)^2 + 1}{(u^y)^2 + 1} \\
(A_x^{\mathrm{IS}})_{25} &= -\frac{(u^\tau)^2}{(u^y)^2 + 1} \\
(A_x^{\mathrm{IS}})_{26} &= \frac{2u^x u^y}{(u^y)^2 + 1} \\
(A_x^{\mathrm{IS}})_{27} &= (u^x)^2 + 1 \\
(A_x^{\mathrm{IS}})_{31} &= u^x u^y (p''(T)T + p'(T)) \\
(A_x^{\mathrm{IS}})_{32} &= u^y (p'(T)T + \Pi) \\
(A_x^{\mathrm{IS}})_{33} &= u^x (p'(T)T + \Pi) \\
(A_x^{\mathrm{IS}})_{36} &= 1 \\
(A_x^{\mathrm{IS}})_{37} &= u^x u^y \\
(A_x^{\mathrm{IS}})_{42} &= -\frac{2 \left(3\pi^{xy} \tau_\pi u^x u^y \left((u^y)^2 + 1 \right) - 3\pi^{yy} \tau_\pi (u^x)^2 (u^y)^2 + (u^\tau)^2 \left((u^y)^2 + 1 \right) \eta \right)}{3(u^\tau)^2} \\
(A_x^{\mathrm{IS}})_{43} &= \frac{2u^x u^y \left(\pi^{xy} \tau_\pi u^x u^y - \pi^{yy} \tau_\pi \left((u^x)^2 + 1 \right) + (u^\tau)^2 \eta \right)}{(u^\tau)^2} \\
(A_x^{\mathrm{IS}})_{44} &= \tau_\pi u^x \\
(A_x^{\mathrm{IS}})_{52} &= -\frac{2\eta}{3\tau^2}
\end{aligned}
$$

$$(A_x^{\mathrm{IS}})_{55} = \frac{\tau_\pi u^x}{\tau^2}$$

$$(A_x^{\mathrm{IS}})_{62} = \frac{u^x \left(u^y \left(3\pi^\eta{}_\eta \tau_\pi \left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi \left(u^x\right)^2 + \tau_\pi\right) + \left(u^\tau\right)^2 \eta\right) - 3\pi^{xy}\tau_\pi u^x \left(2\left(u^y\right)^2 + 1\right)\right)}{3\left(u^\tau\right)^2}$$

$$(A_x^{\mathrm{IS}})_{63} = \frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}\tau_\pi \left(u^x\right)^2\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right) + \pi^{xy}\tau_\pi u^x u^y\left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right)$$
$$- \left(u^\tau\right)^2\left(\pi^\eta{}_\eta \tau_\pi \left(u^x\right)^2\left(u^y\right)^2 - \left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\eta\right)\Big)$$

$$(A_x^{\mathrm{IS}})_{66} = \tau_\pi u^x$$

$$(A_x^{\mathrm{IS}})_{72} = \zeta$$

$$(A_x^{\mathrm{IS}})_{77} = \tau_\Pi u^x$$

$$(A_y^{\mathrm{IS}})_{11} = u^\tau u^y (p''(T)T + p'(T))$$

$$(A_y^{\mathrm{IS}})_{12} = \frac{u^x u^y\left(p'(T)T\left(u^\tau\right)^2 + \Pi\left(u^\tau\right)^2 - \pi^{yy}\right) + \pi^{xy}\left(\left(u^y\right)^2+1\right)}{\left(u^\tau\right)^3}$$

$$(A_y^{\mathrm{IS}})_{13} = \frac{\left(u^\tau\right)^4\left(p'(T)T+\Pi\right) + \left(u^\tau\right)^2\left(u^y\right)^2\left(p'(T)T+\Pi\right) - u^y(\pi^{xy}u^x + \pi^{yy}u^y) + \pi^{yy}\left(u^\tau\right)^2}{\left(u^\tau\right)^3}$$

$$(A_y^{\mathrm{IS}})_{14} = \frac{u^y}{u^\tau}$$

$$(A_y^{\mathrm{IS}})_{16} = \frac{u^x}{u^\tau}$$

$$(A_y^{\mathrm{IS}})_{17} = u^\tau u^y$$

$$(A_y^{\mathrm{IS}})_{21} = u^x u^y (p''(T)T + p'(T))$$

$$(A_y^{\mathrm{IS}})_{22} = u^y (p'(T)T + \Pi)$$

$$(A_y^{\mathrm{IS}})_{23} = u^x (p'(T)T + \Pi)$$

$$(A_y^{\mathrm{IS}})_{26} = 1$$

$$(A_y^{\mathrm{IS}})_{27} = u^x u^y$$

$$(A_y^{\mathrm{IS}})_{31} = p''(T)T\left(u^y\right)^2 + p'(T)\left(u^y\right)^2 + p'(T)$$

$$(A_y^{\mathrm{IS}})_{33} = 2u^y(p'(T)T + \Pi)$$

$$(A_y^{\mathrm{IS}})_{34} = 1$$

$$(A_y^{\mathrm{IS}})_{37} = \left(u^y\right)^2 + 1$$

$$(A_y^{\mathrm{IS}})_{42} = -\frac{2\tau_\pi\left(u^y\right)^2\left(\pi^{xy}\left(u^y\right)^2 + \pi^{xy} - \pi^{yy}u^x u^y\right)}{\left(u^\tau\right)^2}$$

$$(A_y^{\mathrm{IS}})_{43} = \frac{6\pi^{xy}\tau_\pi u^x\left(u^y\right)^3 - 6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)\left(u^y\right)^2 + 4\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\eta}{3\left(u^\tau\right)^2}$$

$$(A_y^{\mathrm{IS}})_{44} = \tau_\pi u^y$$

$$(A_y^{\mathrm{IS}})_{53} = -\frac{2\eta}{3\tau^2}$$

$$(A_y^{\mathrm{IS}})_{55} = \frac{\tau_\pi u^y}{\tau^2}$$

$$(A_y^{\mathrm{IS}})_{62} = \frac{1}{\left(u^\tau\right)^2}\Big(-\pi^{xy}\tau_\pi u^x u^y\left(2\left(u^y\right)^2+1\right) + \eta + \left(u^y\right)^4\left(\pi^\eta{}_\eta\tau_\pi + \eta\right) + \left(u^y\right)^2\left(\pi^\eta{}_\eta\tau_\pi + \pi^{yy}\tau_\pi + 2\eta\right)$$
$$+ \left(u^x\right)^2\left(\left(u^y\right)^2\left(\pi^\eta{}_\eta\tau_\pi + 2\pi^{yy}\tau_\pi + \eta\right) + \eta\right)\Big)$$

$$(A_y^{\mathrm{IS}})_{63} = -\frac{1}{3\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}u^y\Big(3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right) + 3\pi^{xy}\tau_\pi u^y\left(\left(1-2\left(u^x\right)^2\right)\left(u^y\right)^2+1\right)$$
$$+ \left(u^\tau\right)^2 u^x\left(3\pi^\eta{}_\eta\tau_\pi\left(u^y\right)^2 - \left(\left(u^y\right)^2+1\right)\eta\right)\Big)$$

$$(A_y^{\mathrm{IS}})_{66} = \tau_\pi u^y$$

$$(A_y^{\mathrm{IS}})_{73} = \zeta$$
$$(A_y^{\mathrm{IS}})_{77} = \tau_\Pi u^y$$

$$\begin{aligned}
S_1^{\mathrm{IS}} = &\frac{1}{\tau\left(\left(u^y\right)^2+1\right)}\Big(p'(T)T - \pi^{yy}\left(u^x\right)^2 + p'(T)T\left(u^x\right)^2 + 2\pi^{xy}u^xu^y + \pi^{yy}\left(u^y\right)^2 + 2p'(T)T\left(u^y\right)^2 \\
&+ p'(T)T\left(u^x\right)^2\left(u^y\right)^2 + p'(T)T\left(u^y\right)^4 + \Pi\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) + \pi^\eta{}_\eta\left(-\left(u^x\right)^2+\left(u^y\right)^2+1\right)\Big) \\
S_2^{\mathrm{IS}} = &\frac{1}{\tau u^\tau\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}u^x + p'(T)Tu^x - \pi^\eta{}_\eta\left(u^\tau\right)^2u^x - \pi^{yy}\left(u^x\right)^3 + p'(T)T\left(u^x\right)^3 + \pi^{xy}u^y + 2\pi^{xy}\left(u^x\right)^2u^y \\
&+ 2p'(T)Tu^x\left(u^y\right)^2 + p'(T)T\left(u^x\right)^3\left(u^y\right)^2 + \pi^{xy}\left(u^y\right)^3 + p'(T)Tu^x\left(u^y\right)^4 \\
&+ \Pi\left(u^\tau\right)^2u^x\left(\left(u^y\right)^2+1\right)\Big) \\
S_3^{\mathrm{IS}} = &\frac{u^y\left(p'(T)T\left(u^\tau\right)^2 + \Pi\left(u^\tau\right)^2 + \pi^{yy}\right) + \pi^{xy}u^x}{\tau u^\tau} \\
S_4^{\mathrm{IS}} = &\pi^{yy} - \frac{2u^\tau\left(\left(u^y\right)^2+1\right)\eta}{3\tau} \\
S_5^{\mathrm{IS}} = &\frac{3\pi^\eta{}_\eta\tau + 4u^\tau\eta}{3\tau^3} \\
S_6^{\mathrm{IS}} = &\pi^{xy} - \frac{2u^\tau u^xu^y\eta}{3\tau} \\
S_7^{\mathrm{IS}} = &\Pi + \frac{u^\tau\zeta}{\tau}
\end{aligned}$$

1.3.2 $\delta_{\pi\pi}$ part

$$\begin{aligned}
(A_t^{(\delta)})_{42} &= \frac{\pi^{yy}u^x}{u^\tau} \\
(A_t^{(\delta)})_{43} &= \frac{\pi^{yy}u^y}{u^\tau} \\
(A_t^{(\delta)})_{52} &= \frac{\pi^\eta{}_\eta u^x}{\tau^2u^\tau} \\
(A_t^{(\delta)})_{53} &= \frac{\pi^\eta{}_\eta u^y}{\tau^2u^\tau} \\
(A_t^{(\delta)})_{62} &= \frac{\pi^{xy}u^x}{u^\tau} \\
(A_t^{(\delta)})_{63} &= \frac{\pi^{xy}u^y}{u^\tau}
\end{aligned}$$

$$\begin{aligned}
(A_x^{(\delta)})_{42} &= \pi^{yy} \\
(A_x^{(\delta)})_{52} &= \frac{\pi^\eta{}_\eta}{\tau^2} \\
(A_x^{(\delta)})_{62} &= \pi^{xy}
\end{aligned}$$

$$\begin{aligned}
(A_y^{(\delta)})_{43} &= \pi^{yy} \\
(A_y^{(\delta)})_{53} &= \frac{\pi^\eta{}_\eta}{\tau^2} \\
(A_y^{(\delta)})_{63} &= \pi^{xy}
\end{aligned}$$

$$\begin{aligned}
S_4^{(\delta)} &= \frac{\pi^{yy}u^\tau}{\tau} \\
S_5^{(\delta)} &= \frac{\pi^\eta{}_\eta u^\tau}{\tau^3} \\
S_6^{(\delta)} &= \frac{\pi^{xy}u^\tau}{\tau}
\end{aligned}$$

1.3.3 Co-rotating terms (diagnostics)

Not part of the system above; exported alone so that a test can ask whether a 2+1D matrix carries them. The projected vorticity coupling, without a factor τ_π (`two_d_viscous_vorticity`):

$$2\,\pi_\lambda{}^{\langle\mu}\omega^{\nu\rangle\lambda}, \qquad \omega^{\mu\nu} = \tfrac{1}{2}\Delta^{\mu\alpha}\Delta^{\nu\beta}(\nabla_\alpha u_\beta - \nabla_\beta u_\alpha)$$

and the unprojected commutator (`two_d_viscous_notebook_commutator`):

$$\tau_\pi\big(\omega^\mu{}_\lambda\pi^{\lambda\nu} - \pi^{\mu\lambda}\omega_\lambda{}^\nu\big), \qquad \omega_{\mu\nu} = \tfrac{1}{2}(\nabla_\mu u_\nu - \nabla_\nu u_\mu)$$

Both carry no source (V-GV-src, V-GN-src). Nonzero entries of their matrices, $A^{(\omega)}$ for the first and $A^{(\mathrm{c})}$ for the second, rows and columns as above:

$$A_t^{(\omega)} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}, \qquad A_x^{(\omega)} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}, \qquad A_y^{(\omega)} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

$$A_t^{(\mathrm{c})} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}, \qquad A_x^{(\mathrm{c})} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}, \qquad A_y^{(\mathrm{c})} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

1.4 Checks

id	what it tests	
`V-G0cyl`	cylindrical Milne: Nabla g == 0	PASS
`V-G0car`	Cartesian Milne: Nabla g == 0	PASS
`V-G1cyl`	cylindrical: Nabla.V = X' + X/tau	PASS
`V-G1car`	Cartesian: Nabla.V = X' + X/tau	PASS
`V-G2cyl-uu`	u.u == -1	PASS
`V-G2cyl-un`	u_mu pi^{mu nu} == 0	PASS
`V-G2cyl-tr`	pi^mu_mu == 0	PASS
`V-G3cyl`	cylindrical system is quasi-linear	PASS
`V-G6cyl`	cylindrical: (IS + dpp pi theta in eq. 2) == IS + dpp * (the exported dpp part), equation by equation	PASS
`V-G4`	1-D: derived == shipped one_d_viscous_matrix, A_t, A_r, src entry by entry, read from the .jl, SYMBOLIC dpp [CONTROL]	PASS
`V-G4-0`	1-D at dpp = 0 (plain IS): plain IS == the shipped IS part	PASS
`V-G4-1`	1-D at dpp = 4/3 tauS (default)	PASS
`V-G4m`	1-D at dpp = 4/3 tauS == the medium block of the 7-field :MIS one_d_viscous_matrix_fugacity, raw, read from the .jl	PASS
`V-G4m-ctl`	control: at dpp = 0 it is NOT the :MIS medium block	PASS
`V-G2car-uu`	u.u == -1	PASS
`V-G2car-un`	u_mu pi^{mu nu} == 0	PASS
`V-G2car-tr`	pi^mu_mu == 0	PASS
`V-G3car`	Cartesian system is quasi-linear	PASS
`V-G6car`	Cartesian: (IS + dpp pi theta in eq. 2) == IS + dpp * (the exported dpp part), equation by equation	PASS
`V-G5-e`	energy row on Bjorken == d_tau e + (e+p+Pi+pi^eta_eta)/tau	PASS
`V-G5-pi`	pi^eta_eta row on Bjorken == tauS pi' + pi + 4 eta/(3 tau) + dpp pi/tau (theta = 1/tau)	PASS
`V-G5-Pi`	bulk row on Bjorken == tauB Pi' + Pi + zeta/tau	PASS
`V-G5-mom`	the x, y and xy rows vanish identically on Bjorken flow	PASS
`V-GV-anti`	omega is antisymmetric	PASS

id	what it tests	
`V-GV-perp`	omega is transverse to u	PASS
`V-GV-src`	the vorticity term is purely a gradient term (no source)	PASS
`V-GN-anti`	the notebook's omega is antisymmetric	PASS
`V-GN-src`	the notebook's commutator is purely a gradient term (no source)	PASS

28 passed, 0 failed (28 checks evaluated).

Not checked in this script: the slab limit (2+1D at $u^y = 0$, $\partial_y = 0$ against 1+1D at $r \to \infty$); it is a numerical comparison of the two generated routines.

Re-run with `wolframscript -file deriv/derive_2p1d_viscous.wls`.

2 Charm first moment on a 2+1D background ($\nabla\alpha$ closure)

Script: `deriv/derive_hq_2p1d.wls`

Writes: `src/Matrix/HQ_2p1d_BG_generated.jl`

2.1 Setup

Cartesian Milne coordinates, boost invariant, on a prescribed background T, u^x, u^y:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \qquad u^\mu = (u^\tau, u^x, u^y, 0), \quad u^\tau = \sqrt{(u^x)^2 + (u^y)^2 + 1}$$

Charm is a passive tracer:

$$N^\mu = n\, u^\mu + \nu^\mu, \qquad \nabla_\mu N^\mu = 0$$
$$\tau_n\, \Delta^\mu{}_\beta\, D\nu^\beta + \nu^\mu + \kappa\, \Delta^{\mu\beta}\partial_\beta\alpha = 0$$
$$n = n(T, \alpha), \qquad \kappa = D_s\, \partial_\alpha n, \qquad D = u^\mu \nabla_\mu, \qquad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$

Navier–Stokes limit: Fick's law $\nu^\mu = -\kappa\, \nabla_\perp^\mu \alpha$. Fields, with ν^τ from $u_\mu \nu^\mu = 0$:

$$\phi = (\alpha, \nu^x, \nu^y), \qquad \nu^\tau = \frac{\nu^x u^x + \nu^y u^y}{u^\tau}$$

Rows: $\nabla_\mu N^\mu$ and the x, y components of the current equation, in

$$A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0$$

2.2 Result

$$\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n) u^\tau \\
(A_t)_{12} &= \frac{u^x}{u^\tau} \\
(A_t)_{13} &= \frac{u^y}{u^\tau} \\
(A_t)_{21} &= \kappa u^\tau u^x \\
(A_t)_{22} &= \tau_n u^\tau \\
(A_t)_{31} &= \kappa u^\tau u^y \\
(A_t)_{33} &= \tau_n u^\tau
\end{aligned}$$

$$\begin{aligned}
(A_x)_{11} &= (\partial_\alpha n) u^x \\
(A_x)_{12} &= 1 \\
(A_x)_{21} &= \kappa \left((u^x)^2 + 1 \right) \\
(A_x)_{22} &= \tau_n u^x \\
(A_x)_{31} &= \kappa u^x u^y \\
(A_x)_{33} &= \tau_n u^x
\end{aligned}$$

$$\begin{aligned}
(A_y)_{11} &= (\partial_\alpha n) u^y \\
(A_y)_{13} &= 1 \\
(A_y)_{21} &= \kappa u^x u^y \\
(A_y)_{22} &= \tau_n u^y \\
(A_y)_{31} &= \kappa \left((u^y)^2 + 1 \right) \\
(A_y)_{33} &= \tau_n u^y
\end{aligned}$$

$$
\begin{aligned}
S_1 = \frac{1}{\tau\left(u^\tau\right)^3}\Big(&(\partial_\tau u^x)\nu^x\tau\left(u^\tau\right)^2 + (\partial_\tau u^y)\nu^y\tau\left(u^\tau\right)^2 + (\partial_x u^x)n\tau\left(u^\tau\right)^3 + (\partial_y u^y)n\tau\left(u^\tau\right)^3 + n\left(u^\tau\right)^4 \\
&+ (\partial_\tau T)(\partial_T n)\tau\left(u^\tau\right)^4 + \nu^x\left(u^\tau\right)^2 u^x + (\partial_\tau u^x)n\tau\left(u^\tau\right)^2 u^x + (\partial_x T)(\partial_T n)\tau\left(u^\tau\right)^3 u^x \\
&- (\partial_\tau u^x)\nu^x\tau\left(u^x\right)^2 + \nu^y\left(u^\tau\right)^2 u^y + (\partial_\tau u^y)n\tau\left(u^\tau\right)^2 u^y + (\partial_y T)(\partial_T n)\tau\left(u^\tau\right)^3 u^y \\
&- (\partial_\tau u^y)\nu^x\tau u^x u^y - (\partial_\tau u^x)\nu^y\tau u^x u^y - (\partial_\tau u^y)\nu^y\tau\left(u^y\right)^2\Big) \\
S_2 = \frac{1}{\left(u^\tau\right)^2}\Big(&-\nu^y\tau_n u^x\Big((\partial_\tau u^y)u^\tau + (\partial_x u^y)\left(u^x\right)^3 + (\partial_y u^y)u^y + \left(u^x\right)^2((\partial_\tau u^y)u^\tau - (\partial_x u^x)u^y + (\partial_y u^y)u^y) \\
&+ u^x((\partial_x u^y) - u^y((\partial_\tau u^x)u^\tau + (\partial_y u^x)u^y))\Big) \\
&+ \nu^x\Big(1 + (\partial_x u^y)\tau_n\left(u^x\right)^3 u^y + \left(u^y\right)^2 - \tau_n u^x\left(\left(u^y\right)^2 + 1\right)((\partial_\tau u^x)u^\tau + (\partial_y u^x)u^y) \\
&+ \left(u^x\right)^2\left((\partial_\tau u^y)\tau_n u^\tau u^y - (\partial_x u^x)\tau_n\left(\left(u^y\right)^2 + 1\right) + (\partial_y u^y)\tau_n\left(u^y\right)^2 + 1\right)\Big)\Big) \\
S_3 = \frac{1}{\left(u^\tau\right)^2}\Big(&-\nu^x\tau_n u^y\Big((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x + u^y((\partial_y u^x) - u^x((\partial_\tau u^y)u^\tau + (\partial_x u^y)u^x)) \\
&+ \left(u^y\right)^2((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x - (\partial_y u^y)u^x) + (\partial_y u^x)\left(u^y\right)^3\Big) \\
&+ \nu^y\Big(1 - (\partial_\tau u^y)\tau_n u^\tau u^y - (\partial_x u^y)\tau_n\left(u^x\right)^3 u^y + \left(u^y\right)^2(1 - (\partial_y u^y)\tau_n) \\
&+ \left(u^x\right)^2\left(-(\partial_\tau u^y)\tau_n u^\tau u^y + \tau_n\left(u^y\right)^2((\partial_x u^x) - (\partial_y u^y)) + 1\right) \\
&+ \tau_n u^x u^y(u^y((\partial_\tau u^x)u^\tau + (\partial_y u^x)u^y) - (\partial_x u^y))\Big)\Big)
\end{aligned}
$$

2.2.1 Live background

The same three rows with the medium promoted to fields, $\phi_L = (T, u^x, u^y, \alpha, \nu^x, \nu^y)$. Columns 4–6 are the matrices above (H-G6a); columns 1–3 and the remaining source:

$$
\begin{aligned}
(A_t^L)_{11} &= (\partial_T n)u^\tau \\
(A_t^L)_{12} &= \frac{n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^y\right)^2 + 1\right) - \nu^y u^x u^y}{\left(u^\tau\right)^3} \\
(A_t^L)_{13} &= \frac{n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 1\right)}{\left(u^\tau\right)^3} \\
(A_t^L)_{22} &= \frac{\tau_n u^x\left(\nu^y u^x u^y - \nu^x\left(\left(u^y\right)^2 + 1\right)\right)}{u^\tau} \\
(A_t^L)_{23} &= -\frac{\tau_n u^x\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{u^\tau} \\
(A_t^L)_{32} &= -\frac{\tau_n u^y\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{u^\tau} \\
(A_t^L)_{33} &= \frac{\tau_n u^y\left(\nu^x u^x u^y - \nu^y\left(\left(u^x\right)^2 + 1\right)\right)}{u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
(A_x^L)_{11} &= (\partial_T n)u^x \\
(A_x^L)_{12} &= n \\
(A_x^L)_{22} &= \frac{\tau_n\left(u^x\right)^2\left(\nu^y u^x u^y - \nu^x\left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_x^L)_{23} &= -\frac{\tau_n\left(u^x\right)^2\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2}
\end{aligned}
$$

$$(A_x^L)_{32} = \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_x^L)_{33} = \frac{\tau_n u^x u^y \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$

$$(A_y^L)_{11} = (\partial_T n) u^y$$
$$(A_y^L)_{13} = n$$
$$(A_y^L)_{22} = \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{23} = -\frac{\tau_n u^x u^y \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{32} = -\frac{\tau_n \left(u^y\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{33} = \frac{\tau_n \left(u^y\right)^2 \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$

$$S_1^L = \frac{n \left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau}$$
$$S_2^L = \nu^x$$
$$S_3^L = \nu^y$$

2.3 Checks

id	what it tests	
`H-G0cyl`	cylindrical Milne: Nabla g == 0	PASS
`H-G0car`	Cartesian Milne: Nabla g == 0	PASS
`H-G1cyl`	cylindrical: Nabla.V = X' + X/tau for V=(X(tau),0,0,0)	PASS
`H-G1car`	Cartesian: Nabla.V = X' + X/tau	PASS
`H-G2cyl-uu`	u.u == -1	PASS
`H-G2cyl-un`	u.nu == 0	PASS
`H-G3cyl`	cylindrical system is quasi-linear	PASS
`H-G4-At`	cylindrical A_t * u^tau^3 r tau == Fluidum's shipped A_t	PASS
`H-G4-Ax`	cylindrical A_x * u^tau^3 r tau == Fluidum's shipped A_x	PASS
`H-G4-src`	cylindrical src * u^tau^3 r tau == Fluidum's shipped source	PASS
`H-G4src`	cylindrical A_t, A_x, src * u^tau^3 r tau == HQ_const_BG.jl READ FROM THE FILE	PASS
`H-G2car-uu`	u.u == -1	PASS
`H-G2car-un`	u.nu == 0	PASS
`H-G3car`	Cartesian system is quasi-linear	PASS
`H-G5`	slab limit (u^y -> 0, d_y -> 0) decouples nu^y from (alpha, nu^x)	PASS
`H-G6a`	the (alpha, nu^x, nu^y) columns of the 10-field rows == the background-mode matrices	PASS
`H-G6b`	src_live + (medium columns).(background derivatives) == src_background: same system, re-sorted	PASS

17 passed, 0 failed (17 checks evaluated).

Re-run with `wolframscript -file deriv/derive_hq_2p1d.wls`.

3 Consistent charm first moment on a 2+1D background

Script: `deriv/derive_hq_consistent_2p1d.wls`

Writes: `src/Matrix/HQ_2p1d_BG_consistent_generated.jl`

3.1 Setup

Cartesian Milne coordinates, boost invariant, on a prescribed background T, u^x, u^y:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \qquad u^\mu = (u^\tau, u^x, u^y, 0), \quad u^\tau = \sqrt{(u^x)^2 + (u^y)^2 + 1}$$

The first moment is the projected divergence of the charm stress tensor:

$$\nabla_\mu (n\, u^\mu + \nu^\mu) = 0, \qquad \frac{D_s}{T}\, \Delta^\mu{}_\lambda \nabla_\nu T_Q^{\nu\lambda} + \nu^\mu = 0$$
$$T_Q^{\alpha\beta} = (nh - P_0)\, u^\alpha u^\beta + P_0\, \Delta^{\alpha\beta} + h\,(u^\alpha \nu^\beta + \nu^\alpha u^\beta), \qquad P_0 = nT$$
$$n = n(T, \alpha), \quad h = h(T), \qquad \tau_n = D_s h/T, \quad \kappa = D_s\, \partial_\alpha n$$

At rest the current relaxes to

$$\nu^i = -\kappa\, \partial_i \alpha - \frac{D_s}{T}\,(n + T\, \partial_T n)\, \partial_i T$$

Relative to the $\nabla\alpha$ row of the previous section, the current equation gains (HC-C3, HC-C5)

$$\frac{D_s}{T}(n + T\, \partial_T n)\, \nabla_\perp^\mu T + \tau_n n\, a^\mu + \tau_n \Delta^\mu{}_\lambda\, \nu^\sigma \nabla_\sigma u^\lambda + \nu^\mu \Big(\tau_n \theta + \frac{D_s}{T}\, Dh\Big), \qquad a^\mu = Du^\mu$$

Fields and rows as before:

$$\phi = (\alpha, \nu^x, \nu^y), \qquad \nu^\tau = \frac{\nu^x u^x + \nu^y u^y}{u^\tau}$$
$$A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0$$

3.2 Result

$$\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n) u^\tau \\
(A_t)_{12} &= \frac{u^x}{u^\tau} \\
(A_t)_{13} &= \frac{u^y}{u^\tau} \\
(A_t)_{21} &= D_s (\partial_\alpha n) u^\tau u^x \\
(A_t)_{22} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{31} &= D_s (\partial_\alpha n) u^\tau u^y \\
(A_t)_{33} &= \frac{D_s h u^\tau}{T}
\end{aligned}$$

$$\begin{aligned}
(A_x)_{11} &= (\partial_\alpha n) u^x \\
(A_x)_{12} &= 1 \\
(A_x)_{21} &= D_s (\partial_\alpha n) \left((u^x)^2 + 1\right) \\
(A_x)_{22} &= \frac{D_s h u^x}{T} \\
(A_x)_{31} &= D_s (\partial_\alpha n) u^x u^y
\end{aligned}$$

$$(A_x)_{33} = \frac{D_s h u^x}{T}$$

$$\begin{aligned}
(A_y)_{11} &= (\partial_\alpha n) u^y \\
(A_y)_{13} &= 1 \\
(A_y)_{21} &= D_s (\partial_\alpha n) u^x u^y \\
(A_y)_{22} &= \frac{D_s h u^y}{T} \\
(A_y)_{31} &= D_s (\partial_\alpha n) \left((u^y)^2 + 1 \right) \\
(A_y)_{33} &= \frac{D_s h u^y}{T}
\end{aligned}$$

$$\begin{aligned}
S_1 = \frac{1}{\tau (u^\tau)^3} \Big(& (\partial_\tau u^x) \nu^x \tau (u^\tau)^2 + (\partial_\tau u^y) \nu^y \tau (u^\tau)^2 + (\partial_x u^x) n \tau (u^\tau)^3 + (\partial_y u^y) n \tau (u^\tau)^3 + n (u^\tau)^4 \\
& + (\partial_\tau T)(\partial_T n) \tau (u^\tau)^4 + \nu^x (u^\tau)^2 u^x + (\partial_\tau u^x) n \tau (u^\tau)^2 u^x + (\partial_x T)(\partial_T n) \tau (u^\tau)^3 u^x \\
& - (\partial_\tau u^x) \nu^x \tau (u^x)^2 + \nu^y (u^\tau)^2 u^y + (\partial_\tau u^y) n \tau (u^\tau)^2 u^y + (\partial_y T)(\partial_T n) \tau (u^\tau)^3 u^y \\
& - (\partial_\tau u^y) \nu^x \tau u^x u^y - (\partial_\tau u^x) \nu^y \tau u^x u^y - (\partial_\tau u^y) \nu^y \tau (u^y)^2 \Big) \\
S_2 = \frac{1}{T \tau (u^\tau)^3} \Big(& \tau (u^\tau)^2 \Big(D_s (n + (\partial_T n) T) \Big((\partial_\tau T) (u^\tau)^2 u^x + u^\tau \Big((\partial_x T) (u^x)^2 + (\partial_x T) + (\partial_y T) u^x u^y \Big) \Big) \\
& + \nu^x \Big(D_s h'(T) \Big((\partial_\tau T) (u^\tau)^2 + u^\tau ((\partial_x T) u^x + (\partial_y T) u^y) \Big) + T u^\tau \Big) \Big) \\
& + D_s h \Big(\nu^x \Big((u^x)^4 ((\partial_\tau u^y) \tau u^y + 1) \\
& + \Big((u^y)^2 + 1 \Big) \Big((\partial_\tau u^y) \tau u^y + 2 (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + (u^y)^2 + 1 \Big) \\
& + \tau (u^x)^3 \Big(- (\partial_\tau u^x) (u^y)^2 + (\partial_\tau u^x) + (\partial_x u^y) u^\tau u^y \Big) \\
& - \tau u^x \Big((u^y)^2 + 1 \Big) \Big((\partial_\tau u^x) \Big((u^y)^2 - 1 \Big) + (\partial_y u^x) u^\tau u^y \Big) \\
& + (u^x)^2 \Big(2 + (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + (u^y)^2 (- (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + 2) \\
& + (\partial_\tau u^y) \tau u^y \Big((u^y)^2 + 2 \Big) \Big) \Big) \\
& + \tau \Big(n (u^\tau)^2 \Big((\partial_\tau u^x) (u^\tau)^2 + u^\tau ((\partial_x u^x) u^x + (\partial_y u^x) u^y) \Big) \\
& + \nu^y \Big(- (\partial_\tau u^y) (u^\tau)^2 u^x \Big((u^x)^2 + 1 \Big) + (\partial_\tau u^x) (u^\tau)^2 \Big((u^x)^2 + 1 \Big) u^y \\
& + u^\tau \Big(- u^x \Big(u^y \Big(- (\partial_x u^x) (u^x)^2 + (\partial_y u^y) (u^x)^2 + (\partial_y u^y) \Big) + (\partial_x u^y) \Big((u^x)^3 + u^x \Big) \Big) \\
& + (\partial_y u^x) \Big((u^x)^2 + 1 \Big) \Big((u^y)^2 + 1 \Big) \Big) \Big) \Big) \Big) \Big) \\
S_3 = \frac{1}{T \tau (u^\tau)^3} \Big(& \tau (u^\tau)^2 \Big(D_s (n + (\partial_T n) T) \Big((\partial_\tau T) (u^\tau)^2 u^y + u^\tau \Big((\partial_x T) u^x u^y + (\partial_y T) (u^y)^2 + (\partial_y T) \Big) \Big) \\
& + \nu^y \Big(D_s h'(T) \Big((\partial_\tau T) (u^\tau)^2 + u^\tau ((\partial_x T) u^x + (\partial_y T) u^y) \Big) + T u^\tau \Big) \Big) \\
& + D_s h \Big(\nu^y \Big(1 + (\partial_x u^x) \tau u^\tau + 2 (\partial_y u^y) \tau u^\tau + (\partial_\tau u^y) \tau u^y \\
& + (u^y)^2 ((\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + 2) + (\partial_\tau u^y) \tau (u^y)^3 + (u^y)^4 \\
& + (u^x)^4 (1 - (\partial_\tau u^y) \tau u^y) + \tau (u^x)^3 \Big((\partial_\tau u^x) (u^y)^2 + (\partial_\tau u^x) - (\partial_x u^y) u^\tau u^y \Big) \\
& + (u^x)^2 \Big(2 + (\partial_x u^x) \tau u^\tau + 2 (\partial_y u^y) \tau u^\tau + (u^y)^2 ((\partial_x u^x) \tau u^\tau - (\partial_y u^y) \tau u^\tau + 2) \\
& - (\partial_\tau u^y) \tau (u^y)^3 \Big) + \tau u^x \Big((\partial_\tau u^x) \Big((u^y)^2 + 1 \Big)^2 + u^\tau u^y \Big((\partial_y u^x) (u^y)^2 - (\partial_x u^y) \Big) \Big) \Big) \\
& + \tau \Big(n (u^\tau)^2 \Big((\partial_\tau u^y) (u^\tau)^2 + u^\tau ((\partial_x u^y) u^x + (\partial_y u^y) u^y) \Big) \\
& + \nu^x \Big((\partial_\tau u^y) (u^\tau)^2 u^x \Big((u^y)^2 + 1 \Big) - (\partial_\tau u^x) (u^\tau)^2 u^y \Big((u^y)^2 + 1 \Big)
\end{aligned}$$

$$+ u^{\tau}\left((\partial_x u^y)\left((u^x)^2+1\right)\left((u^y)^2+1\right)\right.$$
$$\left.- u^y\left((\partial_x u^x)u^x\left((u^y)^2+1\right)+u^y\left((\partial_y u^x)\,(u^y)^2+(\partial_y u^x)-(\partial_y u^y)u^x u^y\right)\right)\right)\Big)\Big)\Big)$$

3.3 Checks

id	what it tests	
`HC-G0cyl`	cylindrical Milne: Nabla g == 0	PASS
`HC-G0car`	Cartesian Milne: Nabla g == 0	PASS
`HC-G1cyl`	cylindrical: Nabla.V = X' + X/tau	PASS
`HC-G1car`	Cartesian: Nabla.V = X' + X/tau	PASS
`HC-G2cyl-uu`	u.u == -1	PASS
`HC-G2cyl-un`	u.nu == 0	PASS
`HC-G3cyl`	cylindrical consistent system is quasi-linear	PASS
`HC-C1`	cylindrical static nu=0 source == (Ds/T) d_r P0 (== the 1-D hqc_ matrices source)	PASS
`HC-G2car-uu`	u.u == -1	PASS
`HC-G2car-un`	u.nu == 0	PASS
`HC-G3car`	Cartesian consistent system is quasi-linear	PASS
`HC-C2`	Cartesian at-rest NS source == kappa d_x alpha + (Ds/T)(n + T dn_dT) d_x T	PASS
`HC-C3`	consistent - shipped == the four named terms, exactly (x-row)	PASS
`HC-C4`	slab limit (u^y -> 0, d_y -> 0) decouples nu^y	PASS
`HC-C5`	consistent row minus the named terms == the shipped row	PASS

15 passed, 0 failed (15 checks evaluated).

Re-run with `wolframscript -file deriv/derive_hq_consistent_2p1d.wls`.

4 Medium, charm and the two-way coupled system

Script: `deriv/derive_two_way.wls`, with `derive_two_way_gates.wls` and `derive_two_way_export.wls`
Writes: `src/Matrix/two_way_generated.jl`

4.1 Setup

Three charts, boost invariant: 0+1D Bjorken (fields of τ only), 1+1D cylindrical, 2+1D Cartesian Milne:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \quad u^\mu = (1, 0, 0, 0)$$
$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2)$$
$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2)$$

Medium, with the EOS $P(T)$ kept symbolic:

$$T_m^{\mu\nu} = (e + P + \Pi)\, u^\mu u^\nu + (P + \Pi)\, g^{\mu\nu} + \pi^{\mu\nu}, \qquad e = TP'(T) - P(T)$$
$$\tau_\pi\, \Delta^{\mu\nu}_{\alpha\beta} D\pi^{\alpha\beta} + \pi^{\mu\nu} + 2\eta\, \sigma^{\mu\nu} + \delta_{\pi\pi}\, \pi^{\mu\nu}\theta = 0, \qquad \tau_\Pi D\Pi + \Pi + \zeta\theta + \delta_{\Pi\Pi}\Pi\theta = 0$$

(D, θ, $\Delta^{\mu\nu}$, $\Delta^{\mu\nu}_{\alpha\beta}$, $\sigma^{\mu\nu}$ as in the viscous section.) Charm, with $n = n(T, \alpha)$, $h = h(T)$, $P_0 = nT$ and $\delta M^{\mu\nu} = \pi_Q^{\mu\nu} + \Pi_Q \Delta^{\mu\nu}$:

$$\nabla_\mu (n\, u^\mu + \nu^\mu) = 0, \qquad \frac{D_s}{T}\, \Delta^\mu{}_\lambda \nabla_\nu T_Q^{\nu\lambda} + \nu^\mu = 0$$
$$T_Q^{\mu\nu} = (nh - P_0)\, u^\mu u^\nu + P_0\, \Delta^{\mu\nu} + h\, (u^\mu \nu^\nu + \nu^\mu u^\nu) + c\, \delta M^{\mu\nu}$$
$$S^{\lambda\alpha\beta} = I_{31}\, (u^\lambda \Delta^{\alpha\beta} + 2 \text{ perm.}) + \mathcal{A}\, (u^\lambda u^\alpha \nu^\beta + 2 \text{ perm.}) + \mathcal{B}\, (\Delta^{\lambda\alpha}\nu^\beta + 2 \text{ perm.}) + \mathcal{C}\, (u^\lambda \delta M^{\alpha\beta} + 2 \text{ perm.})$$
$$\mathcal{E}^{\mu\nu} = \Delta^\mu{}_\alpha \Delta^\nu{}_\beta \nabla_\lambda S^{\lambda\alpha\beta}: \qquad \frac{D_s}{2T}\Big(\mathcal{E}^{\mu\nu} - \tfrac{1}{3}\Delta^{\mu\nu}\mathcal{E}^\lambda{}_\lambda\Big) + \pi_Q^{\mu\nu} = 0, \qquad \frac{D_s}{2T}\, \tfrac{1}{3}\mathcal{E}^\lambda{}_\lambda + \Pi_Q = 0$$

"2 perm." are the two other placements of the lone index. $I_{31}(T, \alpha)$, $\mathcal{A}(T)$, $\mathcal{B}(T)$, $\mathcal{C}(T)$ stay symbolic (Boltzmann: $I_{31} = nTh$, $\mathcal{A} = m^2 + 5Th$, $\mathcal{B} = Th$, $\mathcal{C} = mK_4/K_3$). $c = 1$ is $c_M = D_s/T$; $c = 0$ drops δM from T_Q. In 2+1D the traceless rows also carry

$$-(1 - w_\omega)\, \tau_M W^{\mu\nu}, \qquad \tau_M = \frac{D_s \mathcal{C}}{2T}, \qquad W^{\mu\nu} = \pi_{Q\lambda}^\mu \omega^{\lambda\nu} + \pi_{Q\lambda}^\nu \omega^{\lambda\mu}, \qquad \omega^{\mu\nu} = \tfrac{1}{2}(\Delta^{\mu\lambda}\nabla_\lambda u^\nu - \Delta^{\nu\lambda}\nabla_\lambda u^\mu)$$

so $w_\omega = 1$ is the covariant row and $w_\omega = 0$ removes the corotation coupling. The five systems:

$$\text{S1}: \ \nabla_\mu T_m^{\mu\nu} = 0, \ \pi = \Pi = 0 \qquad \text{S2}: \ \text{S1} + \text{the } \pi, \Pi \text{ equations}$$
$$\text{S3}: \ \text{charm number} + \text{first moment, background } T, u \qquad \text{S4}: \ \text{S3} + \text{second moment, background } T, u$$
$$\text{S5}: \ \nabla_\mu (T_m + T_Q)^{\mu\nu} = 0, \ \text{the } \pi, \Pi \text{ equations, the charm rows of S4; all fields evolved}$$

Conformal: $P = a_{\mathrm{SB}} T^4$, no bulk. Source on the left, $A_t\, \partial_\tau \phi + A_r\, \partial_r \phi + S = 0$ in 1+1D and $A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0$ in 2+1D.

4.1.1 Fields

Medium, charm first moment, charm second moment, chart by chart:

$$\text{0+1D}: \ (T, \pi^\eta{}_\eta, \Pi), \ (\alpha), \ (\pi_Q{}^\eta{}_\eta, \Pi_Q)$$
$$\text{1+1D}: \ (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi), \ (\alpha, \nu^r), \ (\pi_Q^{\ell\ell}, \pi_Q^{\hat\phi\hat\phi}, \Pi_Q)$$
$$\text{2+1D}: \ (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi), \ (\alpha, \nu^x, \nu^y), \ (\pi_Q^{xx}, \pi_Q^{xy}, \pi_Q^{yy}, \Pi_Q)$$

Rows, in the same order: the fluid rows $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu i}$ ($T_m + T_Q$ in S5); the shear rows $E^{\eta\eta}$ (0+1D), $E^{\phi\phi}$, $E^{\eta\eta}$ (1+1D), E^{yy}, $E^{\eta\eta}$, E^{xy} (2+1D); the bulk row; $\nabla_\mu N^\mu$; the current components; the traceless second-moment rows, components $\hat{\eta}\hat{\eta}$ (0+1D), $\ell\ell$, $\hat{\phi}\hat{\phi}$ (1+1D), xx, xy, yy (2+1D); the trace row.

In 0+1D the tensors are diagonal and $\nu = 0$:

$$\pi^{\mu\nu} = \mathrm{diag}(0, -\frac{\pi^\eta{}_\eta}{2}, -\frac{\pi^\eta{}_\eta}{2}, \frac{\pi^\eta{}_\eta}{\tau^2}), \qquad \pi_Q^{\mu\nu} = \mathrm{diag}(0, -\frac{{\pi_Q}^\eta{}_\eta}{2}, -\frac{{\pi_Q}^\eta{}_\eta}{2}, \frac{{\pi_Q}^\eta{}_\eta}{\tau^2})$$

In 1+1D the medium $\pi^{\mu\nu}$ is as in the viscous section, and the charm stress is spanned by an orthonormal triad transverse to u:

$$\pi_Q^{\mu\nu} = \pi_Q^{\ell\ell}\,\ell^\mu\ell^\nu + \pi_Q^{\hat{\phi}\hat{\phi}}\,\hat{\phi}^\mu\hat{\phi}^\nu - (\pi_Q^{\ell\ell} + \pi_Q^{\hat{\phi}\hat{\phi}})\,\hat{\eta}^\mu\hat{\eta}^\nu$$
$$\ell^\mu = (u^r, u^\tau, 0, 0), \quad \hat{\phi}^\mu = (0, 0, \frac{1}{r}, 0), \quad \hat{\eta}^\mu = (0, 0, 0, \frac{1}{\tau})$$

In 2+1D the medium $\pi^{\mu\nu}$ is as in the viscous section; π_Q^{xx}, π_Q^{xy}, π_Q^{yy} are independent, the τ row follows from $u_\mu \pi_Q^{\mu\nu} = 0$ and

$${\pi_Q}^\eta{}_\eta = -\frac{\pi_Q^{xx}\,(u^y)^2 + \pi_Q^{xx} - 2\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\,(u^x)^2 + \pi_Q^{yy}}{(u^\tau)^2}$$

4.2 Result

Every 0+1D system and the 1+1D conformal systems in full; the generic 1+1D systems and every 2+1D system as the pattern of nonzero entries (•), rows and columns in the field order given. Their entries are in `src/Matrix/two_way_generated.jl`, one function per system.

4.3 0+1D Bjorken

4.3.1 S1, ideal hydro

`ideal_0p1d_generated`, fields $\phi = (T)$.

$$(A_t)_{11} = P''(T)T$$
$$S_1 = \frac{P'(T)T}{\tau}$$

4.3.2 S1, ideal hydro, conformal

`ideal_0p1d_conformal_generated`, fields $\phi = (T)$.

$$(A_t)_{11} = 12a_{\mathrm{SB}}T^3$$
$$S_1 = \frac{4a_{\mathrm{SB}}T^4}{\tau}$$

4.3.3 S2, viscous hydro

`viscous_0p1d_generated`, fields $\phi = (T, \pi^\eta{}_\eta, \Pi)$.

$$(A_t)_{11} = P''(T)T$$
$$(A_t)_{22} = \frac{\tau_\pi}{\tau^2}$$
$$(A_t)_{33} = \tau_\Pi$$
$$S_1 = \frac{P'(T)T + \Pi + \pi^\eta{}_\eta}{\tau}$$

$$
\begin{aligned}
S_2 &= \frac{3\delta_{\pi\pi}\pi^{\eta}{}_{\eta} + 3\pi^{\eta}{}_{\eta}\tau + 4\eta}{3\tau^3} \\
S_3 &= \frac{\delta_{\Pi\Pi}\Pi + \Pi\tau + \zeta}{\tau}
\end{aligned}
$$

4.3.4 S2, viscous hydro, conformal

`viscous_0p1d_conformal_generated`, fields $\phi = (T, \pi^{\eta}{}_{\eta})$.

$$
\begin{aligned}
(A_t)_{11} &= 12 a_{\mathrm{SB}} T^3 \\
(A_t)_{22} &= \frac{\tau_\pi}{\tau^2} \\
S_1 &= \frac{4 a_{\mathrm{SB}} T^4 + \pi^{\eta}{}_{\eta}}{\tau} \\
S_2 &= \frac{3\delta_{\pi\pi}\pi^{\eta}{}_{\eta} + 3\pi^{\eta}{}_{\eta}\tau + 4\eta}{3\tau^3}
\end{aligned}
$$

4.3.5 S3, charm first moment on a background

`hq_fm_bg_0p1d_generated`, fields $\phi = (\alpha)$.

$$
\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n) \\
S_1 &= (\partial_T n)(\partial_\tau T) + \frac{n}{\tau}
\end{aligned}
$$

4.3.6 S4, charm first and second moment on a background

`hq_m2_bg_0p1d_generated`, fields $\phi = (\alpha, \pi_Q{}^{\eta}{}_{\eta}, \Pi_Q)$.

$$
\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n) \\
(A_t)_{22} &= \frac{\mathcal{C} D_s}{2T} \\
(A_t)_{31} &= \frac{(\partial_\alpha I_{31}) D_s}{2T} \\
(A_t)_{33} &= \frac{\mathcal{C} D_s}{2T} \\
S_1 &= (\partial_T n)(\partial_\tau T) + \frac{n}{\tau} \\
S_2 &= \frac{D_s(4\mathcal{C}\Pi_Q + 7\mathcal{C}\pi_Q{}^{\eta}{}_{\eta} + 3\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^{\eta}{}_{\eta}\tau + 4 I_{31})}{6T\tau} + \pi_Q{}^{\eta}{}_{\eta} \\
S_3 &= \frac{D_s(\mathcal{C}(5\Pi_Q + 2\pi_Q{}^{\eta}{}_{\eta}) + 3(\partial_\tau T)\tau(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})) + 5 I_{31})}{6T\tau} + \Pi_Q
\end{aligned}
$$

4.3.7 S5, two-way

`two_way_0p1d_generated`, fields $\phi = (T, \pi^{\eta}{}_{\eta}, \Pi, \alpha, \pi_Q{}^{\eta}{}_{\eta}, \Pi_Q)$.

$$
\begin{aligned}
(A_t)_{11} &= (h'(T) - 1)n + (\partial_T n)(h - T) + P''(T)T \\
(A_t)_{14} &= (\partial_\alpha n)(h - T) \\
(A_t)_{22} &= \frac{\tau_\pi}{\tau^2} \\
(A_t)_{33} &= \tau_\Pi \\
(A_t)_{41} &= (\partial_T n) \\
(A_t)_{44} &= (\partial_\alpha n)
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{51} &= \frac{\mathcal{C}'(T) D_s {\pi_Q}^{\eta}{}_{\eta}}{2T} \\
(A_t)_{55} &= \frac{\mathcal{C} D_s}{2T} \\
(A_t)_{61} &= \frac{D_s(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31}))}{2T} \\
(A_t)_{64} &= \frac{(\partial_\alpha I_{31}) D_s}{2T} \\
(A_t)_{66} &= \frac{\mathcal{C} D_s}{2T} \\
S_1 &= \frac{c\Pi_Q + c{\pi_Q}^{\eta}{}_{\eta} + P'(T)T + hn + \Pi + {\pi}^{\eta}{}_{\eta}}{\tau} \\
S_2 &= \frac{3\delta_{\pi\pi}{\pi}^{\eta}{}_{\eta} + 3{\pi}^{\eta}{}_{\eta}\tau + 4\eta}{3\tau^3} \\
S_3 &= \frac{\delta_{\Pi\Pi}\Pi + \Pi\tau + \zeta}{\tau} \\
S_4 &= \frac{n}{\tau} \\
S_5 &= \frac{4\mathcal{C} D_s \Pi_Q + 7\mathcal{C} D_s {\pi_Q}^{\eta}{}_{\eta} + 4 D_s I_{31} + 6{\pi_Q}^{\eta}{}_{\eta} T\tau}{6T\tau} \\
S_6 &= \frac{5\mathcal{C} D_s \Pi_Q + 2\mathcal{C} D_s {\pi_Q}^{\eta}{}_{\eta} + 5 D_s I_{31} + 6\Pi_Q T\tau}{6T\tau}
\end{aligned}
$$

4.3.8 S5, two-way, conformal

`two_way_0p1d_conformal_generated`, fields $\phi = (T, {\pi}^{\eta}{}_{\eta}, \alpha, {\pi_Q}^{\eta}{}_{\eta}, \Pi_Q)$.

$$
\begin{aligned}
(A_t)_{11} &= 12 a_{\mathrm{SB}} T^3 + (h'(T) - 1)n + (\partial_T n)(h - T) \\
(A_t)_{13} &= (\partial_\alpha n)(h - T) \\
(A_t)_{22} &= \frac{\tau_\pi}{\tau^2} \\
(A_t)_{31} &= (\partial_T n) \\
(A_t)_{33} &= (\partial_\alpha n) \\
(A_t)_{41} &= \frac{\mathcal{C}'(T) D_s {\pi_Q}^{\eta}{}_{\eta}}{2T} \\
(A_t)_{44} &= \frac{\mathcal{C} D_s}{2T} \\
(A_t)_{51} &= \frac{D_s(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31}))}{2T} \\
(A_t)_{53} &= \frac{(\partial_\alpha I_{31}) D_s}{2T} \\
(A_t)_{55} &= \frac{\mathcal{C} D_s}{2T} \\
S_1 &= \frac{4 a_{\mathrm{SB}} T^4 + c\Pi_Q + c{\pi_Q}^{\eta}{}_{\eta} + hn + {\pi}^{\eta}{}_{\eta}}{\tau} \\
S_2 &= \frac{3\delta_{\pi\pi}{\pi}^{\eta}{}_{\eta} + 3{\pi}^{\eta}{}_{\eta}\tau + 4\eta}{3\tau^3} \\
S_3 &= \frac{n}{\tau} \\
S_4 &= \frac{4\mathcal{C} D_s \Pi_Q + 7\mathcal{C} D_s {\pi_Q}^{\eta}{}_{\eta} + 4 D_s I_{31} + 6{\pi_Q}^{\eta}{}_{\eta} T\tau}{6T\tau} \\
S_5 &= \frac{5\mathcal{C} D_s \Pi_Q + 2\mathcal{C} D_s {\pi_Q}^{\eta}{}_{\eta} + 5 D_s I_{31} + 6\Pi_Q T\tau}{6T\tau}
\end{aligned}
$$

4.4 1+1D cylindrical

4.4.1 S1, ideal hydro

`ideal_1p1d_generated`, fields $\phi = (T, u^r)$.

$$A_t = \begin{pmatrix} \bullet & \bullet \\ \bullet & \bullet \end{pmatrix}, \qquad A_r = \begin{pmatrix} \bullet & \bullet \\ \bullet & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \end{pmatrix}$$

4.4.2 S1, ideal hydro, conformal

`ideal_1p1d_conformal_generated`, fields $\phi = (T, u^r)$.

$$\begin{aligned}
(A_t)_{11} &= 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2+3\right)\\
(A_t)_{12} &= 8a_{\mathrm{SB}}T^4u^r\\
(A_t)_{21} &= 16a_{\mathrm{SB}}T^3u^ru^\tau\\
(A_t)_{22} &= \frac{4a_{\mathrm{SB}}T^4\left(2\left(u^r\right)^2+1\right)}{u^\tau}\\
(A_r)_{11} &= 16a_{\mathrm{SB}}T^3u^ru^\tau\\
(A_r)_{12} &= \frac{4a_{\mathrm{SB}}T^4\left(2\left(u^r\right)^2+1\right)}{u^\tau}\\
(A_r)_{21} &= 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2+1\right)\\
(A_r)_{22} &= 8a_{\mathrm{SB}}T^4u^r\\
S_1 &= \frac{4a_{\mathrm{SB}}T^4\left(r\left(u^r\right)^2+r+\tau u^ru^\tau\right)}{r\tau}\\
S_2 &= \frac{4a_{\mathrm{SB}}T^4u^r(ru^\tau+\tau u^r)}{r\tau}
\end{aligned}$$

4.4.3 S2, viscous hydro

`viscous_1p1d_generated`, fields $\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_r = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.4.4 S2, viscous hydro, conformal

`viscous_1p1d_conformal_generated`, fields $\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta)$.

$$\begin{aligned}
(A_t)_{11} &= 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2+3\right)\\
(A_t)_{12} &= -2u^r\left(-4a_{\mathrm{SB}}T^4+\pi^\eta{}_\eta+\pi^\phi{}_\phi\right)\\
(A_t)_{13} &= -\left(u^r\right)^2\\
(A_t)_{14} &= -\left(u^r\right)^2\\
(A_t)_{21} &= 16a_{\mathrm{SB}}T^3u^ru^\tau\\
(A_t)_{22} &= -\frac{\left(2\left(u^r\right)^2+1\right)\left(-4a_{\mathrm{SB}}T^4+\pi^\eta{}_\eta+\pi^\phi{}_\phi\right)}{u^\tau}\\
(A_t)_{23} &= -u^ru^\tau
\end{aligned}$$

$$
\begin{aligned}
(A_t)_{24} &= -u^r u^\tau \\
(A_t)_{32} &= \frac{u^r(3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta)}{3r^2u^\tau} \\
(A_t)_{33} &= \frac{\tau_\pi u^\tau}{r^2} \\
(A_t)_{42} &= \frac{u^r(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2u^\tau} \\
(A_t)_{44} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_r)_{11} &= 16a_{\rm SB}T^3u^ru^\tau \\
(A_r)_{12} &= -\frac{\left(2\left(u^r\right)^2+1\right)\left(-4a_{\rm SB}T^4+\pi^\eta{}_\eta+\pi^\phi{}_\phi\right)}{u^\tau} \\
(A_r)_{13} &= -u^ru^\tau \\
(A_r)_{14} &= -u^ru^\tau \\
(A_r)_{21} &= 4a_{\rm SB}T^3\left(4\left(u^r\right)^2+1\right) \\
(A_r)_{22} &= -2u^r\left(-4a_{\rm SB}T^4+\pi^\eta{}_\eta+\pi^\phi{}_\phi\right) \\
(A_r)_{23} &= -\left(u^r\right)^2-1 \\
(A_r)_{24} &= -\left(u^r\right)^2-1 \\
(A_r)_{32} &= \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta}{3r^2} \\
(A_r)_{33} &= \frac{\tau_\pi u^r}{r^2} \\
(A_r)_{42} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2} \\
(A_r)_{44} &= \frac{\tau_\pi u^r}{\tau^2} \\
S_1 &= \frac{1}{r\tau u^\tau}\Big(4a_{\rm SB}T^4\left(\left(u^r\right)^2+1\right)\left(ru^\tau+\tau u^r\right)-\pi^\phi{}_\phi u^r\left(ru^ru^\tau+\tau\left(u^r\right)^2+\tau\right) \\
&\quad -\pi^\eta{}_\eta\left(r\left(\left(u^r\right)^2-1\right)u^\tau+\tau\left(\left(u^r\right)^3+u^r\right)\right)\Big) \\
S_2 &= \frac{1}{r\tau u^\tau}\Big(-\pi^\eta{}_\eta\left(\left(u^r\right)^2+1\right)\left(ru^r+\tau u^\tau\right)+4a_{\rm SB}T^4u^r\left(r\left(u^r\right)^2+r+\tau u^ru^\tau\right) \\
&\quad -\pi^\phi{}_\phi\left(r\left(\left(u^r\right)^3+u^r\right)+\tau\left(\left(u^r\right)^2+2\right)u^\tau\right)\Big) \\
S_3 &= \frac{3\pi^\phi{}_\phi(r(\delta_{\pi\pi}u^\tau+\tau)+\delta_{\pi\pi}\tau u^r)-2ru^\tau\eta+4\tau u^r\eta}{3r^3\tau} \\
S_4 &= \frac{3\pi^\eta{}_\eta(r(\delta_{\pi\pi}u^\tau+\tau)+\delta_{\pi\pi}\tau u^r)+4ru^\tau\eta-2\tau u^r\eta}{3r\tau^3}
\end{aligned}
$$

4.4.5 S3, charm first moment on a background

`hq_fm_bg_1p1d_generated`, fields $\phi = (\alpha, \nu^r)$.

$$
A_t = \begin{pmatrix} \bullet & \bullet \\ \bullet & \bullet \end{pmatrix}, \qquad A_r = \begin{pmatrix} \bullet & \bullet \\ \bullet & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \end{pmatrix}
$$

4.4.6 S4, charm first and second moment on a background

`hq_m2_bg_1p1d_generated`, fields $\phi = (\alpha, \nu^r, \pi_Q^{\ell\ell}, \pi_Q^{\hat\phi\hat\phi}, \Pi_Q)$.

$$
A_t = \begin{pmatrix} \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_r = \begin{pmatrix} \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

4.4.7 S5, two-way

`two_way_1p1d_generated`, fields $\phi = (T, u^r, \pi^{\phi}{}_{\phi}, \pi^{\eta}{}_{\eta}, \Pi, \alpha, \nu^r, \pi_Q^{\ell\ell}, \pi_Q^{\hat{\phi}\hat{\phi}}, \Pi_Q)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_r = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}$$

$$S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.4.8 S5, two-way, conformal

`two_way_1p1d_conformal_generated`, fields $\phi = (T, u^r, \pi^{\phi}{}_{\phi}, \pi^{\eta}{}_{\eta}, \alpha, \nu^r, \pi_Q^{\ell\ell}, \pi_Q^{\hat{\phi}\hat{\phi}}, \Pi_Q)$.

$$\begin{aligned}
(A_t)_{11} &= 2h'(T)\nu^r u^r + 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2 + 3\right) + n\left(h'(T)\left(u^r\right)^2 + h'(T) - 1\right) \\
&\quad + (\partial_T n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{12} &= 2\left(u^r\left(4a_{\mathrm{SB}}T^4 + c\Pi_Q + c\pi_Q^{\ell\ell} - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}\right) + h(nu^r + \nu^r)\right) \\
(A_t)_{13} &= -\left(u^r\right)^2 \\
(A_t)_{14} &= -\left(u^r\right)^2 \\
(A_t)_{15} &= (\partial_\alpha n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{16} &= 2hu^r \\
(A_t)_{17} &= c\left(u^r\right)^2 \\
(A_t)_{19} &= c\left(u^r\right)^2 \\
(A_t)_{21} &= \frac{u^r\left(\left(u^r\right)^2 + 1\right)\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(n\left(\left(u^r\right)^2 + 1\right)u^r + 2\nu^r\left(u^r\right)^2 + \nu^r\right)}{u^\tau} \\
(A_t)_{22} &= \frac{1}{\left(u^\tau\right)^3}\Big(-\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\left(-4a_{\mathrm{SB}}T^4 - c\Pi_Q - c\pi_Q^{\ell\ell} + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}\right) \\
&\quad + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big) \\
(A_t)_{23} &= -u^r u^\tau \\
(A_t)_{24} &= -u^r u^\tau \\
(A_t)_{25} &= (\partial_\alpha n)hu^r u^\tau \\
(A_t)_{26} &= \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau} \\
(A_t)_{27} &= cu^r u^\tau \\
(A_t)_{29} &= cu^r u^\tau
\end{aligned}$$

$$(A_t)_{32} = \frac{u^r(3\delta_{\pi\pi}\pi^{\phi}{}_{\phi} - 2\eta)}{3r^2u^\tau}$$
$$(A_t)_{33} = \frac{\tau_\pi u^\tau}{r^2}$$
$$(A_t)_{42} = \frac{u^r(3\delta_{\pi\pi}\pi^{\eta}{}_{\eta} - 2\eta)}{3\tau^2u^\tau}$$
$$(A_t)_{44} = \frac{\tau_\pi u^\tau}{\tau^2}$$
$$(A_t)_{51} = (\partial_T n)u^\tau$$
$$(A_t)_{52} = \frac{nu^r\left((u^r)^2+1\right)+\nu^r}{(u^\tau)^3}$$
$$(A_t)_{55} = (\partial_\alpha n)u^\tau$$
$$(A_t)_{56} = \frac{u^r}{u^\tau}$$
$$(A_t)_{61} = \frac{D_s u^\tau(h'(T)\nu^r + u^r((\partial_T n)T+n))}{T}$$
$$(A_t)_{62} = \frac{D_s\left(c\left((u^r)^2+1\right)(\Pi_Q+\pi_Q^{\ell\ell})+h\left(n(u^r)^2+n+\nu^r u^r\right)\right)}{Tu^\tau}$$
$$(A_t)_{65} = (\partial_\alpha n)D_s u^r u^\tau$$
$$(A_t)_{66} = \frac{D_s h u^\tau}{T}$$
$$(A_t)_{67} = \frac{cD_s u^r u^\tau}{T}$$
$$(A_t)_{69} = \frac{cD_s u^r u^\tau}{T}$$
$$(A_t)_{71} = \frac{D_s\left(4\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\pi_Q^{\ell\ell}\left((u^r)^2+1\right)\right)}{6Tu^\tau}$$
$$(A_t)_{72} = \frac{D_s\left(4\mathcal{A}\nu^r\left((u^r)^2+1\right)+4\mathcal{B}\nu^r+u^r\left((u^r)^2+1\right)(4\mathcal{C}\Pi_Q+7\mathcal{C}\pi_Q^{\ell\ell}+4I_{31})\right)}{6T(u^\tau)^3}$$
$$(A_t)_{76} = \frac{2\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{77} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{81} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q^{\hat\phi\hat\phi}\left((u^r)^2+1\right)-2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau}$$
$$(A_t)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r\left((u^r)^2+1\right)+2\mathcal{B}\nu^r+u^r\left((u^r)^2+1\right)(2\mathcal{C}\Pi_Q+2\mathcal{C}\pi_Q^{\ell\ell}-3\mathcal{C}\pi_Q^{\hat\phi\hat\phi}+2I_{31})\right)}{6T(u^\tau)^3}$$
$$(A_t)_{86} = -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{88} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{91} = \frac{D_s\left(5\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\Pi_Q\left((u^r)^2+1\right)+3(\partial_T I_{31})\left((u^r)^2+1\right)\right)}{6Tu^\tau}$$
$$(A_t)_{92} = \frac{D_s\left(2\mathcal{A}\nu^r\left((u^r)^2+1\right)+5\mathcal{B}\nu^r+u^r\left((u^r)^2+1\right)(5\mathcal{C}\Pi_Q+2\mathcal{C}\pi_Q^{\ell\ell}+5I_{31})\right)}{6T(u^\tau)^3}$$
$$(A_t)_{95} = \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T}$$
$$(A_t)_{96} = \frac{5\mathcal{B}D_s u^r}{6Tu^\tau}$$
$$(A_t)_{99} = \frac{\mathcal{C}D_s u^\tau}{2T}$$

$$
\begin{aligned}
(A_r)_{11} &= \frac{u^r\left(\left(u^r\right)^2+1\right)\left(16a_{\rm SB}T^3+(\partial_T n)h\right)+h'(T)\left(n\left(\left(u^r\right)^2+1\right)u^r+2\nu^r\left(u^r\right)^2+\nu^r\right)}{u^\tau}\\
(A_r)_{12} &= \frac{1}{\left(u^\tau\right)^3}\Big(-\left(2\left(u^r\right)^4+3\left(u^r\right)^2+1\right)\left(-4a_{\rm SB}T^4-c\Pi_Q-c\pi^{\ell\ell}_Q+\pi^\eta{}_\eta+\pi^\phi{}_\phi\right)\\
&\quad +h\left(2n\left(u^r\right)^4+3n\left(u^r\right)^2+n+2\nu^r\left(u^r\right)^3+3\nu^r u^r\right)\Big)\\
(A_r)_{13} &= -u^r u^\tau\\
(A_r)_{14} &= -u^r u^\tau\\
(A_r)_{15} &= (\partial_\alpha n)hu^r u^\tau\\
(A_r)_{16} &= \frac{h\left(2\left(u^r\right)^2+1\right)}{u^\tau}\\
(A_r)_{17} &= cu^r u^\tau\\
(A_r)_{19} &= cu^r u^\tau\\
(A_r)_{21} &= 16a_{\rm SB}T^3\left(u^r\right)^2+4a_{\rm SB}T^3+h'(T)n\left(u^r\right)^2+2h'(T)\nu^r u^r+(\partial_T n)h\left(u^r\right)^2+(\partial_T n)T+n\\
(A_r)_{22} &= 2\left(u^r\left(4a_{\rm SB}T^4+c\Pi_Q+c\pi^{\ell\ell}_Q-\pi^\eta{}_\eta-\pi^\phi{}_\phi\right)+h(nu^r+\nu^r)\right)\\
(A_r)_{23} &= -\left(u^r\right)^2-1\\
(A_r)_{24} &= -\left(u^r\right)^2-1\\
(A_r)_{25} &= (\partial_\alpha n)\left(h\left(u^r\right)^2+T\right)\\
(A_r)_{26} &= 2hu^r\\
(A_r)_{27} &= c\left(\left(u^r\right)^2+1\right)\\
(A_r)_{29} &= c\left(\left(u^r\right)^2+1\right)\\
(A_r)_{32} &= \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi-2\eta}{3r^2}\\
(A_r)_{33} &= \frac{\tau_\pi u^r}{r^2}\\
(A_r)_{42} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta}{3\tau^2}\\
(A_r)_{44} &= \frac{\tau_\pi u^r}{\tau^2}\\
(A_r)_{51} &= (\partial_T n)u^r\\
(A_r)_{52} &= n\\
(A_r)_{55} &= (\partial_\alpha n)u^r\\
(A_r)_{56} &= 1\\
(A_r)_{61} &= \frac{D_s\left(h'(T)\nu^r u^r+(\partial_T n)T\left(\left(u^r\right)^2+1\right)+n\left(\left(u^r\right)^2+1\right)\right)}{T}\\
(A_r)_{62} &= \frac{D_s\left(cu^r(\Pi_Q+\pi^{\ell\ell}_Q)+hnu^r+\frac{h\nu^r\left(\left(u^r\right)^2+2\right)}{\left(u^\tau\right)^2}\right)}{T}\\
(A_r)_{65} &= (\partial_\alpha n)D_s\left(\left(u^r\right)^2+1\right)\\
(A_r)_{66} &= \frac{D_s hu^r}{T}\\
(A_r)_{67} &= \frac{cD_s\left(\left(u^r\right)^2+1\right)}{T}\\
(A_r)_{69} &= \frac{cD_s\left(\left(u^r\right)^2+1\right)}{T}\\
(A_r)_{71} &= \frac{D_s(4\mathcal{B}'(T)\nu^r+3\mathcal{C}'(T)\pi^{\ell\ell}_Q u^r)}{6T}
\end{aligned}
$$

$$(A_r)_{72} = \frac{D_s\left(4\mathcal{A}\nu^r u^r + \mathcal{C}\left(\left(u^r\right)^2+1\right)\left(4\Pi_Q + 7\pi_Q^{\ell\ell}\right) + 4I_{31}\left(\left(u^r\right)^2+1\right)\right)}{6T\left(u^\tau\right)^2}$$

$$(A_r)_{76} = \frac{2\mathcal{B}D_s}{3T}$$

$$(A_r)_{77} = \frac{\mathcal{C}D_s u^r}{2T}$$

$$(A_r)_{81} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q^{\hat{\phi}\hat{\phi}}u^r)}{6T}$$

$$(A_r)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(\left(u^r\right)^2+1\right)\left(2\Pi_Q + 2\pi_Q^{\ell\ell} - 3\pi_Q^{\hat{\phi}\hat{\phi}}\right) + 2I_{31}\left(\left(u^r\right)^2+1\right)\right)}{6T\left(u^\tau\right)^2}$$

$$(A_r)_{86} = -\frac{\mathcal{B}D_s}{3T}$$

$$(A_r)_{88} = \frac{\mathcal{C}D_s u^r}{2T}$$

$$(A_r)_{91} = \frac{D_s(5\mathcal{B}'(T)\nu^r + 3u^r(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}$$

$$(A_r)_{92} = \frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(\left(u^r\right)^2+1\right)\left(5\Pi_Q + 2\pi_Q^{\ell\ell}\right) + 5I_{31}\left(\left(u^r\right)^2+1\right)\right)}{6T\left(u^\tau\right)^2}$$

$$(A_r)_{95} = \frac{(\partial_\alpha I_{31})D_s u^r}{2T}$$

$$(A_r)_{96} = \frac{5\mathcal{B}D_s}{6T}$$

$$(A_r)_{99} = \frac{\mathcal{C}D_s u^r}{2T}$$

$$\begin{aligned}
S_1 = \frac{1}{r\tau u^\tau}\Big(&-\pi^\phi{}_\phi\tau u^r + c\Pi_Q\tau u^r + c\pi_Q^{\ell\ell}\tau u^r + 4a_{\mathrm{SB}}T^4\tau u^r - \pi^\phi{}_\phi\tau\left(u^r\right)^3 + c\Pi_Q\tau\left(u^r\right)^3 + c\pi_Q^{\ell\ell}\tau\left(u^r\right)^3 \\
&+ 4a_{\mathrm{SB}}T^4\tau\left(u^r\right)^3 + c\Pi_Q r u^\tau - c\pi_Q^{\ell\ell}ru^\tau - c\pi_Q^{\hat{\phi}\hat{\phi}}ru^\tau + 4a_{\mathrm{SB}}rT^4u^\tau - \pi^\phi{}_\phi r\left(u^r\right)^2u^\tau + c\Pi_Q r\left(u^r\right)^2u^\tau \\
&+ c\pi_Q^{\ell\ell}r\left(u^r\right)^2u^\tau + 4a_{\mathrm{SB}}rT^4\left(u^r\right)^2u^\tau - \pi^\eta{}_\eta\left(r\left(\left(u^r\right)^2-1\right)u^\tau + \tau\left(\left(u^r\right)^3+u^r\right)\right) \\
&+ h\left(n\left(\left(u^r\right)^2+1\right)\left(ru^\tau+\tau u^r\right) + \nu^r\left(2ru^ru^\tau + 2\tau\left(u^r\right)^2+\tau\right)\right)\Big)
\end{aligned}$$

$$\begin{aligned}
S_2 = \frac{1}{r\tau u^\tau}\Big(&-\pi^\phi{}_\phi ru^r + c\Pi_Q ru^r + c\pi_Q^{\ell\ell}ru^r + 4a_{\mathrm{SB}}rT^4u^r - \pi^\phi{}_\phi r\left(u^r\right)^3 + c\Pi_Q r\left(u^r\right)^3 + c\pi_Q^{\ell\ell}r\left(u^r\right)^3 \\
&+ 4a_{\mathrm{SB}}rT^4\left(u^r\right)^3 - 2\pi^\phi{}_\phi\tau u^\tau + c\pi_Q^{\ell\ell}\tau u^\tau - c\pi_Q^{\hat{\phi}\hat{\phi}}\tau u^\tau - \pi^\phi{}_\phi\tau\left(u^r\right)^2u^\tau + c\Pi_Q\tau\left(u^r\right)^2u^\tau \\
&+ c\pi_Q^{\ell\ell}\tau\left(u^r\right)^2u^\tau + 4a_{\mathrm{SB}}T^4\tau\left(u^r\right)^2u^\tau - \pi^\eta{}_\eta\left(\left(u^r\right)^2+1\right)\left(ru^r+\tau u^\tau\right) \\
&+ hnu^r\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right) + h\nu^r\left(2r\left(u^r\right)^2 + r + 2\tau u^ru^\tau\right)\Big)
\end{aligned}$$

$$S_3 = \frac{3\pi^\phi{}_\phi(r(\delta_{\pi\pi}u^\tau+\tau) + \delta_{\pi\pi}\tau u^r) - 2ru^\tau\eta + 4\tau u^r\eta}{3r^3\tau}$$

$$S_4 = \frac{3\pi^\eta{}_\eta(r(\delta_{\pi\pi}u^\tau+\tau) + \delta_{\pi\pi}\tau u^r) + 4ru^\tau\eta - 2\tau u^r\eta}{3r\tau^3}$$

$$S_5 = \frac{nu^r+\nu^r}{r} + \frac{n\left(u^r\right)^2 + n + \nu^ru^r}{\tau u^\tau}$$

$$\begin{aligned}
S_6 = \frac{1}{rT\tau u^\tau}\Big(&\nu^r rT\tau u^\tau + cD_s\left(\left(u^r\right)^2+1\right)\left(2\pi_Q^{\ell\ell}ru^r + \pi_Q^{\ell\ell}\tau u^\tau + \pi_Q^{\hat{\phi}\hat{\phi}}ru^r - \pi_Q^{\hat{\phi}\hat{\phi}}\tau u^\tau\right) \\
&+ D_s h\nu^r\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right)\Big)
\end{aligned}$$

$$\begin{aligned}
S_7 = \pi_Q^{\ell\ell} &- \frac{1}{6rT\tau u^\tau}D_s\Big(2\mathcal{B}\nu^r(ru^r+\tau u^\tau) + 2I_{31}\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right) \\
&+ \mathcal{C}\Big(-5\pi_Q^{\ell\ell}r\left(\left(u^r\right)^2+1\right) - 3\pi_Q^{\ell\ell}\tau u^ru^\tau - 2\pi_Q^{\hat{\phi}\hat{\phi}}\left(r\left(u^r\right)^2 + r - \tau u^ru^\tau\right) \\
&+ 2\Pi_Q\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right)\Big)\Big)
\end{aligned}$$

$$
\begin{aligned}
S_8 &= \pi_Q^{\hat{\phi}\hat{\phi}} \\
&\quad + \frac{1}{6rT\tau u^\tau} D_s \Big(-2\mathcal{B}\nu^r (ru^r - 2\tau u^\tau) - 2I_{31} \left(r\left(u^r\right)^2 + r - 2\tau u^r u^\tau \right) \\
&\quad + \mathcal{C}\Big(2\pi_Q^{\ell\ell} r \left(\left(u^r\right)^2 + 1 \right) + 5\pi_Q^{\hat{\phi}\hat{\phi}} r \left(\left(u^r\right)^2 + 1 \right) + 7\pi_Q^{\hat{\phi}\hat{\phi}} \tau u^r u^\tau \\
&\quad - 2\Pi_Q \left(r\left(u^r\right)^2 + r - 2\tau u^r u^\tau \right) \Big) \Big) \\
S_9 &= \Pi_Q \\
&\quad + \frac{1}{6rT\tau u^\tau} D_s \Big(5\mathcal{B}\nu^r (ru^r + \tau u^\tau) + 5I_{31} \left(r\left(u^r\right)^2 + r + \tau u^r u^\tau \right) + 5\mathcal{C}\Pi_Q \left(r\left(u^r\right)^2 + r + \tau u^r u^\tau \right) \\
&\quad - 2\mathcal{C} \left(\pi_Q^{\ell\ell} r \left(\left(u^r\right)^2 + 1 \right) + \pi_Q^{\hat{\phi}\hat{\phi}} \left(r\left(u^r\right)^2 + r - \tau u^r u^\tau \right) \right) \Big)
\end{aligned}
$$

4.5 2+1D Cartesian

4.5.1 S1, ideal hydro

`ideal_2p1d_generated`, fields $\phi = (T, u^x, u^y)$.

$$
A_t = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \bullet & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

4.5.2 S1, ideal hydro, conformal

`ideal_2p1d_conformal_generated`, fields $\phi = (T, u^x, u^y)$.

$$
A_t = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \bullet & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

4.5.3 S2, viscous hydro

`viscous_2p1d_generated`, fields $\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi)$.

$$
A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet \\ \bullet & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}
$$

$$
S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

4.5.4 S2, viscous hydro, conformal

`viscous_2p1d_conformal_generated`, fields $\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy})$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet \\ \bullet & \cdot & \bullet & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \bullet & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet \end{pmatrix}$$

$$S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.5.5 S3, charm first moment on a background

`hq_fm_bg_2p1d_generated`, fields $\phi = (\alpha, \nu^x, \nu^y)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \cdot \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.5.6 S4, charm first and second moment on a background

`hq_m2_bg_2p1d_generated`, fields $\phi = (\alpha, \nu^x, \nu^y, \pi_Q^{xx}, \pi_Q^{xy}, \pi_Q^{yy}, \Pi_Q)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \bullet & \bullet & \cdot & \bullet \\ \bullet & \cdot & \bullet & \cdot & \bullet & \bullet & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet \\ \bullet & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet \\ \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \bullet \\ \bullet & \cdot & \bullet & \cdot & \cdot & \bullet & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}$$

$$S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.5.7 S5, two-way

`two_way_2p1d_generated`, fields $\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y, \pi_Q^{xx}, \pi_Q^{xy}, \pi_Q^{yy}, \Pi_Q)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \cdot & \bullet \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}$$

$$A_y = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \cdot & \bullet \\
\bullet & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet & \bullet \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.5.8 S5, two-way, conformal

`two_way_2p1d_conformal_generated`, fields $\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \alpha, \nu^x, \nu^y, \pi_Q^{xx}, \pi_Q^{xy}, \pi_Q^{yy}, \Pi_Q)$.

$$A_t = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\
\bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet & \bullet & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad A_x = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \cdot & \bullet \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet & \cdot & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet
\end{pmatrix}$$

$$A_y = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \cdot & \bullet \\
\bullet & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet & \bullet \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet & \bullet \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

4.6 Checks

id	what it tests	
`G0-0p1d`	Nabla g == 0	PASS
`G1-0p1d`	trap detector: Nabla.(X(tau,..),0,0,0) == X' + X/tau (Christoffel AND sqrt(-g) routes)	PASS
`G2-0p1d`	u.u=-1; u.pi=0, tr pi=0 (medium); u.nu=0; u.piQ=0, tr piQ=0; both symmetric	PASS
`G-aQ-0p1d`	aQ' has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
`G3-0p1d`	every row (medium, charm-in-T, first moment, second moment) is quasi-linear	PASS

id	what it tests	
S3-free-0p1d	S3 rows carry no second-moment field and no c (c enters only through dM)	PASS
S34-nomedpi-0p1d	charm rows never see the medium pi or Pi (so S3/S4 need only T, u as background)	PASS
Gc-extract-0p1d	A_t dphi + A_x dphi (+A_y dphi) + src == the raw rows (symbolic)	PASS
Gc-ctrl-0p1d	control: the exact random-point zero test REJECTS a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-0p1d	S5 fluid rows == Nabla_mu (T_m + T_Q)^{mu nu}, sqrt(-g) route (symbolic)	PASS
Gc-number-0p1d	charm row == Nabla.N, sqrt(-g) route (symbolic)	PASS
Gc-c0-0p1d	c = 0: dM columns of the fluid + first-moment rows vanish identically, and their sources carry no dM	PASS
Gc-c1-0p1d	c = 1: the dM columns are LIVE (the gate above is not vacuous) [0+1D: dM enters only the energy row's source]	PASS
Gc-c1src-0p1d	0+1D, c = 1: dM enters the energy row's source (Bjorken work of the charm stress)	PASS
Gc-off-0p1d	charm off (n, nu, dM -> 0): S5's medium block and medium sources = = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 IS S2	PASS
G0-1p1d	Nabla g == 0	PASS
G1-1p1d	trap detector: Nabla.(X(tau,..),0,0,0) == X' + X/tau (Christoffel AND sqrt(-g) routes)	PASS
G2-1p1d	u.u=-1; u.pi=0, tr pi=0 (medium); u.nu=0; u.piQ=0, tr piQ=0; both symmetric	PASS
G2-triad	the (l, phihat, etahat) triad is orthonormal and transverse	PASS
G-aQ-1p1d	aQ' has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
G3-1p1d	every row (medium, charm-in-T, first moment, second moment) is quasi-linear	PASS
S3-free-1p1d	S3 rows carry no second-moment field and no c (c enters only through dM)	PASS
S34-nomedpi-1p1d	charm rows never see the medium pi or Pi (so S3/S4 need only T, u as background)	PASS
Gc-extract-1p1d	A_t dphi + A_x dphi (+A_y dphi) + src == the raw rows (symbolic)	PASS
Gc-ctrl-1p1d	control: the exact random-point zero test REJECTS a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-1p1d	S5 fluid rows == Nabla_mu (T_m + T_Q)^{mu nu}, sqrt(-g) route (symbolic)	PASS
Gc-number-1p1d	charm row == Nabla.N, sqrt(-g) route (symbolic)	PASS
Gc-c0-1p1d	c = 0: dM columns of the fluid + first-moment rows vanish identically, and their sources carry no dM	PASS
Gc-c1-1p1d	c = 1: the dM columns are LIVE (the gate above is not vacuous)	PASS
Gc-off-1p1d	charm off (n, nu, dM -> 0): S5's medium block and medium sources = = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 IS S2	PASS
G0-2p1d	Nabla g == 0	PASS
G1-2p1d	trap detector: Nabla.(X(tau,..),0,0,0) == X' + X/tau (Christoffel AND sqrt(-g) routes)	PASS
G2-2p1d	u.u=-1; u.pi=0, tr pi=0 (medium); u.nu=0; u.piQ=0, tr piQ=0; both symmetric	PASS
G-aQ-2p1d	aQ' has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
G3-2p1d	every row (medium, charm-in-T, first moment, second moment) is quasi-linear	PASS
S3-free-2p1d	S3 rows carry no second-moment field and no c (c enters only through dM)	PASS
S34-nomedpi-2p1d	charm rows never see the medium pi or Pi (so S3/S4 need only T, u as background)	PASS
Gc-extract-2p1d	A_t dphi + A_x dphi (+A_y dphi) + src == the raw rows (exact at 4 random rational points)	PASS
Gc-ctrl-2p1d	control: the exact random-point zero test REJECTS a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-2p1d	S5 fluid rows == Nabla_mu (T_m + T_Q)^{mu nu}, sqrt(-g) route (symbolic)	PASS

id	what it tests	
Gc-number-2p1d	charm row == Nabla.N, sqrt(-g) route (symbolic)	PASS
Gc-c0-2p1d	c = 0: dM columns of the fluid + first-moment rows vanish identically, and their sources carry no dM	PASS
Gc-c1-2p1d	c = 1: the dM columns are LIVE (the gate above is not vacuous)	PASS
Gc-off-2p1d	charm off (n, nu, dM -> 0): S5's medium block and medium sources = = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 IS S2	PASS
R0-jl2wl	Julia -> WL conversion of the shipped/generated matrices (0)	PASS
R-S2-1p1d-shipped	S2 1+1D generic EOS, dpp = dPP = 0 == SHIPPED one_d_viscous_ matrix (1d_viscous_cilindrical.jl), entry by entry (55 entries, symbolic)	PASS
R-S2-1p1d-generator	S2 1+1D, dpp = dPP = 0 == one_d_viscous_matrix_derived (derive_2p1d_viscous.wls output), entry by entry (55 entries, symbolic)	PASS
R-S2-2p1d-generator	S2 2+1D, dpp = dPP = 0 == two_d_viscous_matrix_derived (derive_2p1d_viscous.wls output), entry by entry (154 entries, symbolic)	PASS
R-S3-2p1d-generator	S3 2+1D == hq2d_matrices_consistent (derive_hq_consistent_ 2p1d.wls output), entry by entry (30 entries, symbolic)	PASS
R-S2-1p1d-MIS	S2 1+1D with dpp = 4/3 tauS, dPP = 0 == medium block of one_d_ viscous_matrix_fugacity (:MIS, read from the .jl), entry by entry (55 entries, symbolic)	PASS
R-S2-1p1d-MIS-ctl	control: dpp = 0 does NOT reproduce the :MIS medium block (the two 1-D media differ by delta_pipi)	PASS
R-S4S3-1p1d	S4 rows 1-2 at c = 0 == S3 (so the first-moment block of S4/S5 is the gated consistent closure)	PASS
R-S4-2p1d-fluidum	S4 2+1D second-moment rows == Fluidum hq2d_m2_rows (projector = true), Julia, 24 states: wv=0 <-> vorticity=false: 4.220039818717633e-15, wv=1 <-> vorticity=true: 4.551222145397426e-15	PASS
X-2to1-1	S5 2+1D axisymmetric on the x axis: tau-momentum (symbolic)	PASS
X-2to1-2	S5 2+1D axisymmetric on the x axis: x-momentum == r-momentum (symbolic)	PASS
X-2to1-3	S5 2+1D axisymmetric on the x axis: y-momentum == 0 (symbolic)	PASS
X-2to1-4	S5 2+1D axisymmetric on the x axis: E^{yy} == r^2 E^{phiphi} (symbolic)	PASS
X-2to1-5	S5 2+1D axisymmetric on the x axis: E^{etaeta} (symbolic)	PASS
X-2to1-6	S5 2+1D axisymmetric on the x axis: E^{xy} == 0 (symbolic)	PASS
X-2to1-7	S5 2+1D axisymmetric on the x axis: bulk (symbolic)	PASS
X-2to1-8	S5 2+1D axisymmetric on the x axis: charm number (symbolic)	PASS
X-2to1-9	S5 2+1D axisymmetric on the x axis: nu^x row == nu^r row (symbolic)	PASS
X-2to1-10	S5 2+1D axisymmetric on the x axis: nu^y row == 0 (symbolic)	PASS
X-2to1-11	S5 2+1D axisymmetric on the x axis: piQ^{xx} row == (u^tau)^2 x (l l) row (symbolic)	PASS
X-2to1-12	S5 2+1D axisymmetric on the x axis: piQ^{xy} row == 0 (symbolic)	PASS
X-2to1-13	S5 2+1D axisymmetric on the x axis: piQ^{yy} row == (phi phi) row (symbolic)	PASS
X-2to1-14	S5 2+1D axisymmetric on the x axis: trace row (symbolic)	PASS
X-ctrl	control: the same test REJECTS a wrong pairing (2+1D tau row vs 1+1D r row)	PASS
X-vort-axis	the corotation coupling W vanishes identically on an axisymmetric state (so G-chart is blind to wv, as documented)	PASS
X-1to0-1	S5 1+1D at u^r = 0, no r-dependence: energy (symbolic)	PASS
X-1to0-2	S5 1+1D at u^r = 0, no r-dependence: r-momentum == 0 (symbolic)	PASS
X-1to0-3	S5 1+1D at u^r = 0, no r-dependence: E^{phiphi} == -(tau^2/2r^2) E^{etaeta} (symbolic)	PASS
X-1to0-4	S5 1+1D at u^r = 0, no r-dependence: E^{etaeta} (symbolic)	PASS
X-1to0-5	S5 1+1D at u^r = 0, no r-dependence: bulk (symbolic)	PASS
X-1to0-6	S5 1+1D at u^r = 0, no r-dependence: charm number (symbolic)	PASS
X-1to0-7	S5 1+1D at u^r = 0, no r-dependence: nu^r row == 0 (symbolic)	PASS
X-1to0-8	S5 1+1D at u^r = 0, no r-dependence: (l l) row == -1/2 (eta eta) row (symbolic)	PASS
X-1to0-9	S5 1+1D at u^r = 0, no r-dependence: (phi phi) row == -1/2 (eta eta) row (symbolic)	PASS

id	what it tests	
X-1to0-10	S5 1+1D at u^r = 0, no r-dependence: trace row (symbolic)	PASS
B-ideal-conf	conformal ideal Bjorken: dT/dtau == -T/(3 tau)	PASS
B-ideal-gen	generic-EOS ideal Bjorken: dT/dtau == -P'/(P" tau) == -c_s^2 T/ tau, c_s^2 = P'/(T P")	PASS
B-IS-gen	generic IS Bjorken: T' = -(T P' + Pi + pi)/(tau T P"), tauS pi' = -pi - 4 visc/(3 tau) - dpp pi/tau, tauB Pi' = -Pi - zeta/tau - dPP Pi/tau	PASS
B-IS-conf	conformal IS Bjorken: dT/dtau = -T/(3tau) - pi/(12 a T^3 tau), tauS dpi/dtau = -pi - 4 visc/(3 tau) - dpp pi/tau [dpp = 4/3 tauS: the DNMR delta_pipi form; no tau_pipi, lambda_1]	PASS
B-NS-conf	first-order (NS) limit pi = -4 visc/(3 tau): dT/dtau = -T/(3tau) + (4/ 9)(eta/s)/tau^2, s = 4 a T^3	PASS
B-twoway-energy	S5 Bjorken energy row: d_tau(e + e_Q) + (e+P+Pi+pi + n h + c(piQ^eta_eta + Pi_Q))/tau	PASS
B-charm-number	Bjorken charm number: (1/tau) d_tau (n tau) = 0	PASS
T-rest-1p1d	nu = 0, dM = 0, grad alpha = 0, medium at rest: transverse momentum exchange Delta.Nabla.T_Q == 0 and the nu row is satisfied (c symbolic)	PASS
T-drag-1p1d	drag: on shell, Nabla.T_Q^i = -(T/Ds) nu^i (drag on charm) and the fluid row then gives Nabla.T_m^i = +(T/Ds) nu^i (Newton III), exact, c symbolic	PASS
T-split-1p1d	fluid momentum rows == Nabla.T_m + Nabla.T_Q identically	PASS
T-rest-2p1d	nu = 0, dM = 0, grad alpha = 0, medium at rest: transverse momentum exchange Delta.Nabla.T_Q == 0 and the nu row is satisfied (c symbolic)	PASS
T-drag-2p1d	drag: on shell, Nabla.T_Q^i = -(T/Ds) nu^i (drag on charm) and the fluid row then gives Nabla.T_m^i = +(T/Ds) nu^i (Newton III), exact, c symbolic	PASS
T-split-2p1d	fluid momentum rows == Nabla.T_m + Nabla.T_Q identically	PASS
E-same-ideal_0p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-ideal_0p1d_generated	every free symbol is an argument or a field	PASS
E-same-ideal_0p1d_conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-ideal_0p1d_ conformal_generated	every free symbol is an argument or a field	PASS
E-same-viscous_0p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-viscous_0p1d_ generated	every free symbol is an argument or a field	PASS
E-same-viscous_0p1d_ conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-viscous_0p1d_ conformal_generated	every free symbol is an argument or a field	PASS
E-same-hq_fm_bg_0p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-hq_fm_bg_0p1d_ generated	every free symbol is an argument or a field	PASS
E-same-hq_m2_bg_0p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-hq_m2_bg_0p1d_ generated	every free symbol is an argument or a field	PASS
E-same-two_way_0p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-two_way_0p1d_ generated	every free symbol is an argument or a field	PASS
E-same-two_way_0p1d_ conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-two_way_0p1d_ conformal_generated	every free symbol is an argument or a field	PASS
E-same-ideal_1p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-ideal_1p1d_generated	every free symbol is an argument or a field	PASS
E-same-ideal_1p1d_conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-ideal_1p1d_ conformal_generated	every free symbol is an argument or a field	PASS
E-same-viscous_1p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-viscous_1p1d_ generated	every free symbol is an argument or a field	PASS
E-same-viscous_1p1d_ conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-viscous_1p1d_ conformal_generated	every free symbol is an argument or a field	PASS

id	what it tests	
E-same-hq_fm_bg_1p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-hq_fm_bg_1p1d_generated	every free symbol is an argument or a field	PASS
E-same-hq_m2_bg_1p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-hq_m2_bg_1p1d_generated	every free symbol is an argument or a field	PASS
E-same-two_way_1p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-two_way_1p1d_generated	every free symbol is an argument or a field	PASS
E-same-two_way_1p1d_conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-two_way_1p1d_conformal_generated	every free symbol is an argument or a field	PASS
E-same-ideal_2p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-ideal_2p1d_generated	every free symbol is an argument or a field	PASS
E-same-ideal_2p1d_conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-ideal_2p1d_conformal_generated	every free symbol is an argument or a field	PASS
E-same-viscous_2p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-viscous_2p1d_generated	every free symbol is an argument or a field	PASS
E-same-viscous_2p1d_conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-viscous_2p1d_conformal_generated	every free symbol is an argument or a field	PASS
E-same-hq_fm_bg_2p1d	simplified exported entries == derived entries (symbolic)	PASS
E-free-hq_fm_bg_2p1d_generated	every free symbol is an argument or a field	PASS
E-same-hq_m2_bg_2p1d	simplified exported entries == derived entries (exact at 3 random rational points)	PASS
E-free-hq_m2_bg_2p1d_generated	every free symbol is an argument or a field	PASS
E-same-two_way_2p1d	simplified exported entries == derived entries (exact at 3 random rational points)	PASS
E-free-two_way_2p1d_generated	every free symbol is an argument or a field	PASS
E-same-two_way_2p1d_conformal	simplified exported entries == derived entries (exact at 3 random rational points)	PASS
E-free-two_way_2p1d_conformal_generated	every free symbol is an argument or a field	PASS
E-jl	the generated Julia (24 functions) parses, runs, and reproduces its 24-state CSVs: max rel err 1.6653345369377348e-15	PASS

141 passed, 0 failed (141 checks evaluated). Skipped: `R-S3-1p1d-generator`, `R-S4-1p1d-generator`, `R-S4c-1p1d-generator`. These compare with reference CSVs of the 1+1D derivations of the consistent first and second moment and of the c_M coupling, which are read only when `FLUIDUM_EXTERNAL_REFPOINTS` names a directory holding them; it was unset.

Not a check: the characteristic speeds of S5 are computed and reported (`two_way_char_speeds.tsv` in the scratch directory).

Re-run with `wolframscript -file deriv/derive_two_way.wls`.